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Lin

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(54) **METHOD, USER EQUIPMENT, AND
NON-TRANSITORY COMPUTER READABLE
MEDIUM FOR PHYSICAL LAYER-BASED
PRIORITIZATION WITHOUT LOGICAL
CHANNEL-BASED PRIORITIZATION**

(58) **Field of Classification Search**

None

See application file for complete search history.

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U.S.C. 154(b) by 616 days.

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(57) **ABSTRACT**

A method, a UE, and a non-transitory computer readable medium for PHI-based prioritization without LCH-based prioritization are provided. In an embodiment, the method includes when the LCH-based prioritization is not configured, and the PHY-based prioritization is configured in the UE, receiving, by a PHY layer of the UE, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively; when uplink transmission for the uplink grants is not on-going, delivering, by the PHY layer, first information to a MAC layer of the UE, wherein the first information is for resolving the overlapping conflict by comparing the first PHY priority with the second PHY priority, and generating, by the MAC layer, only a first MAC protocol data unit for a first one of the uplink grants using the first information.

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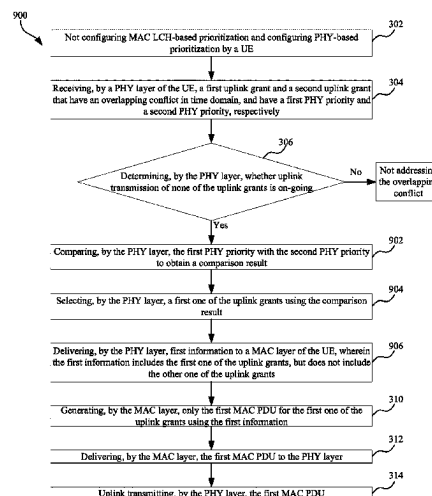
(60) Provisional application No. 63/062,179, filed on Aug. 6, 2020.

(51) **Int. Cl.**
H04W 72/23 (2023.01)

H04W 72/566 (2023.01)

(52) **U.S. Cl.**
CPC **H04W 72/23** (2023.01); **H04W 72/569**
(2023.01)

17 Claims, 12 Drawing Sheets



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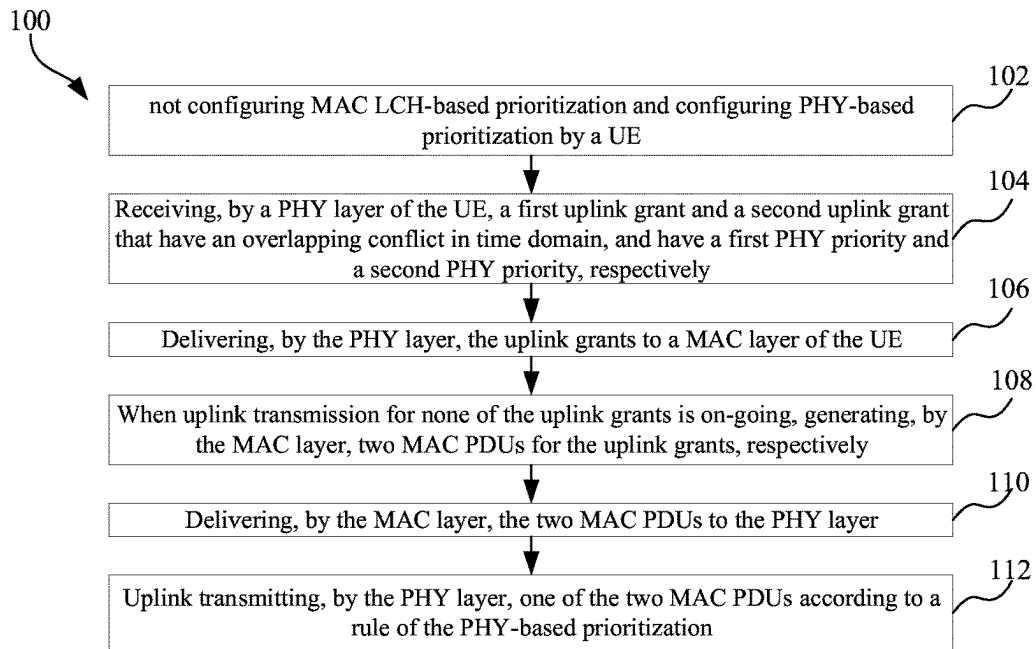


FIG. 1

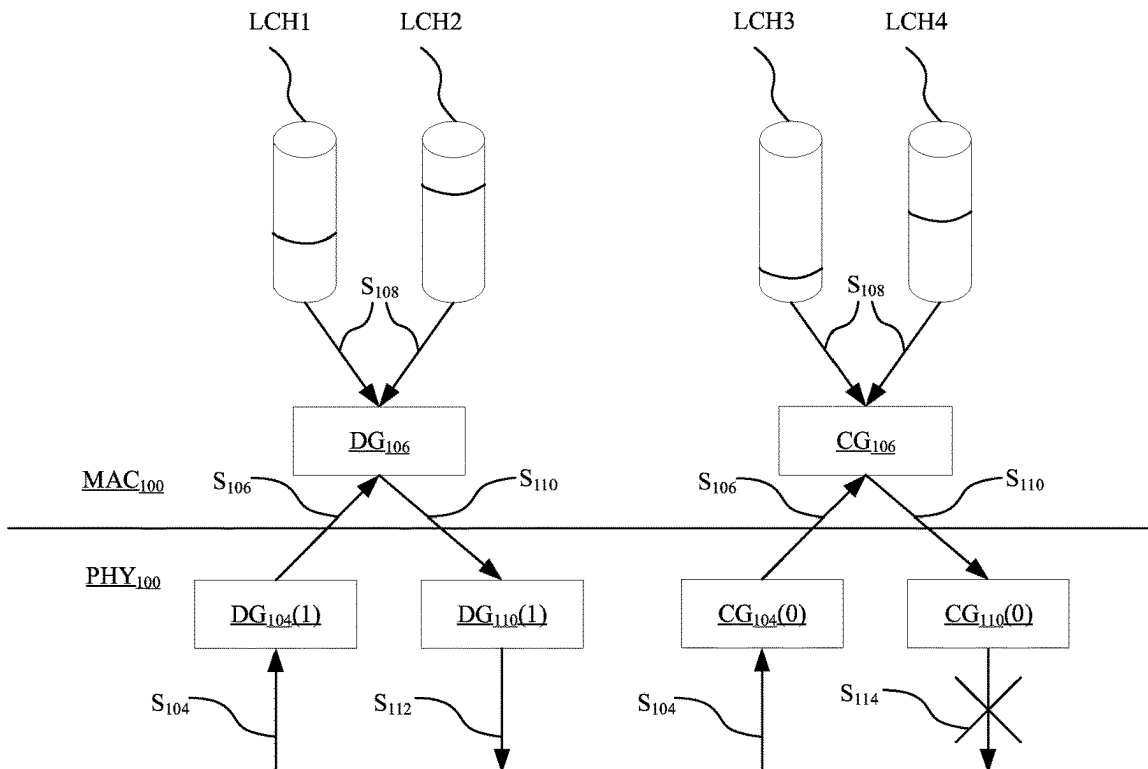


FIG. 2

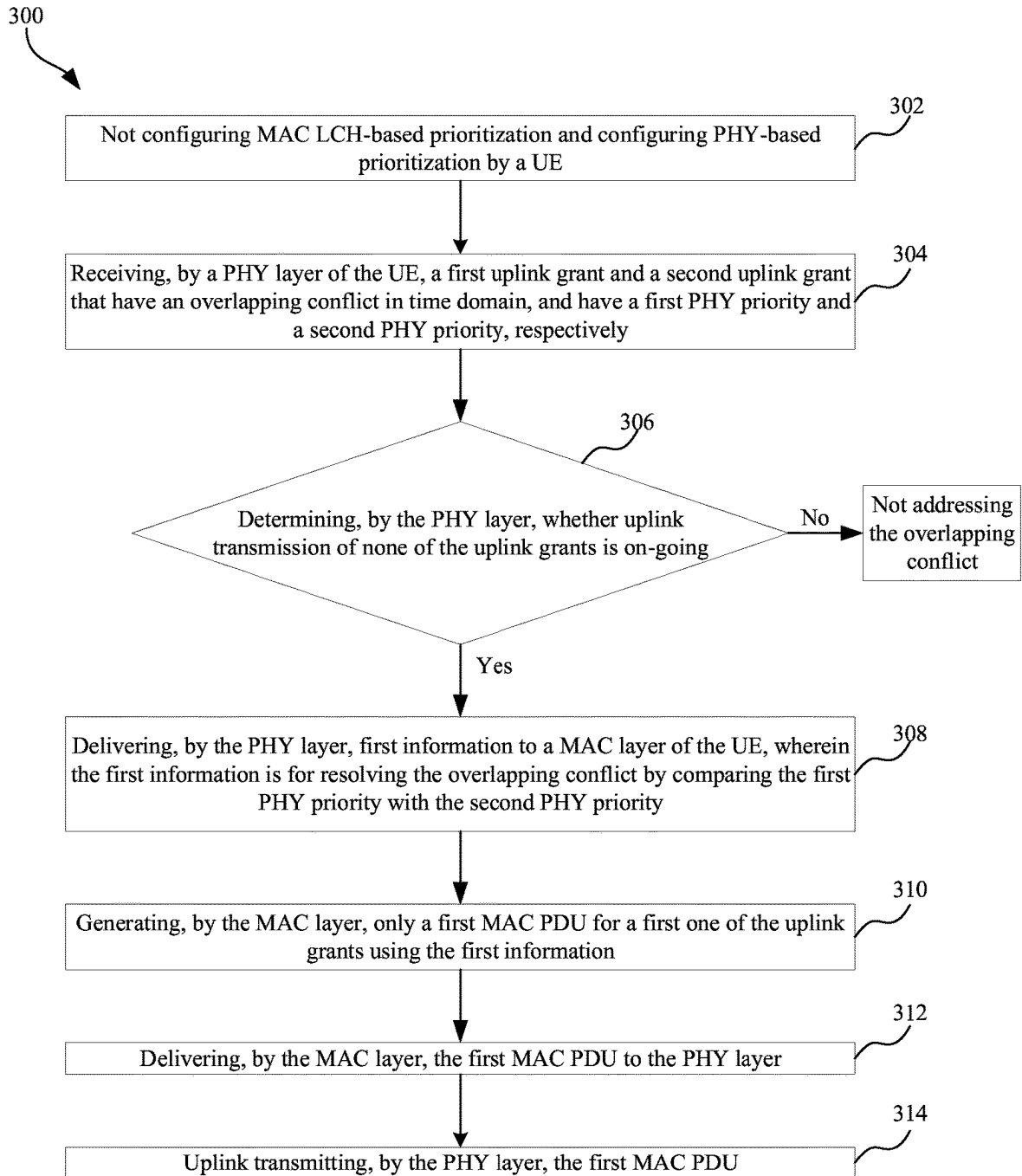


FIG. 3

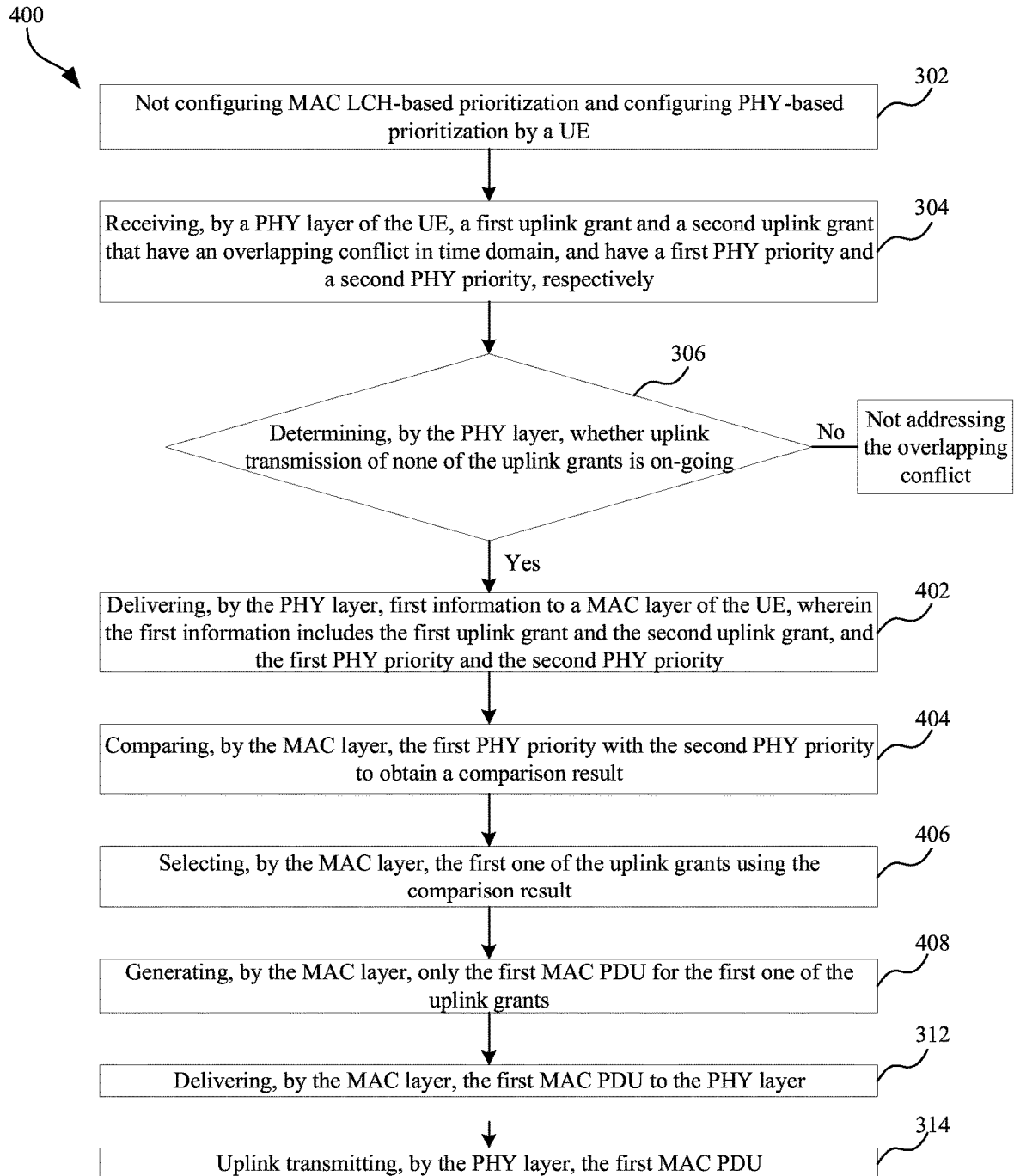


FIG. 4

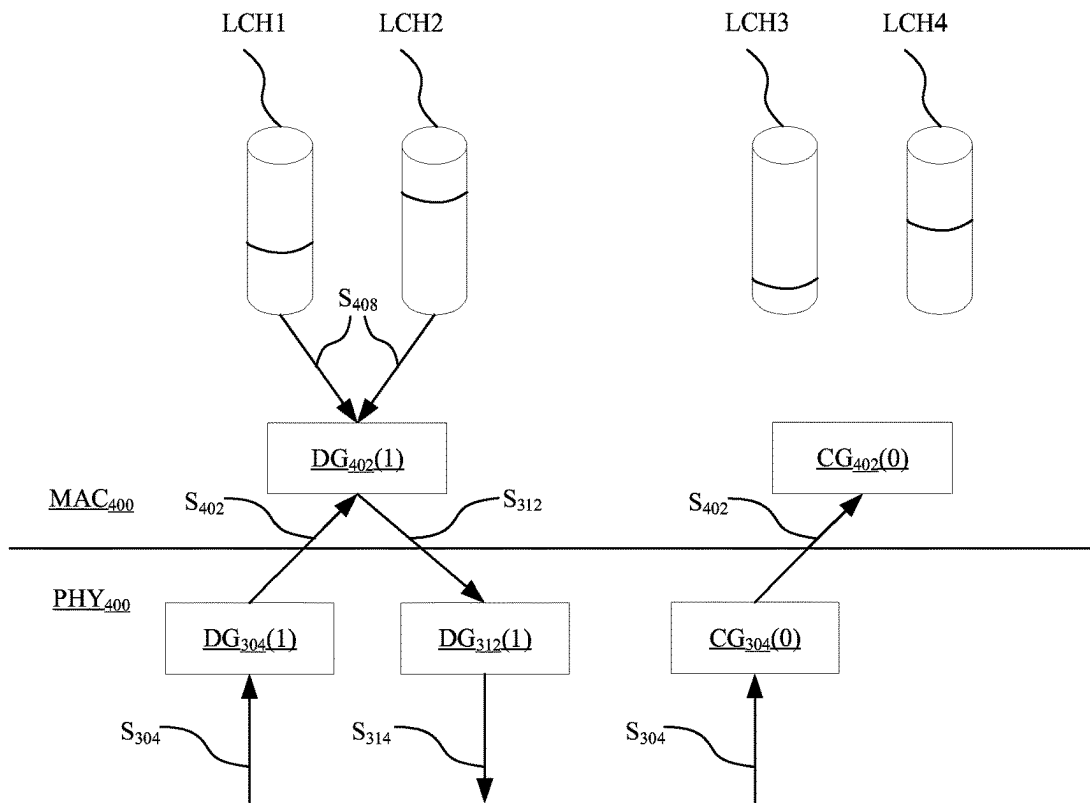


FIG. 5

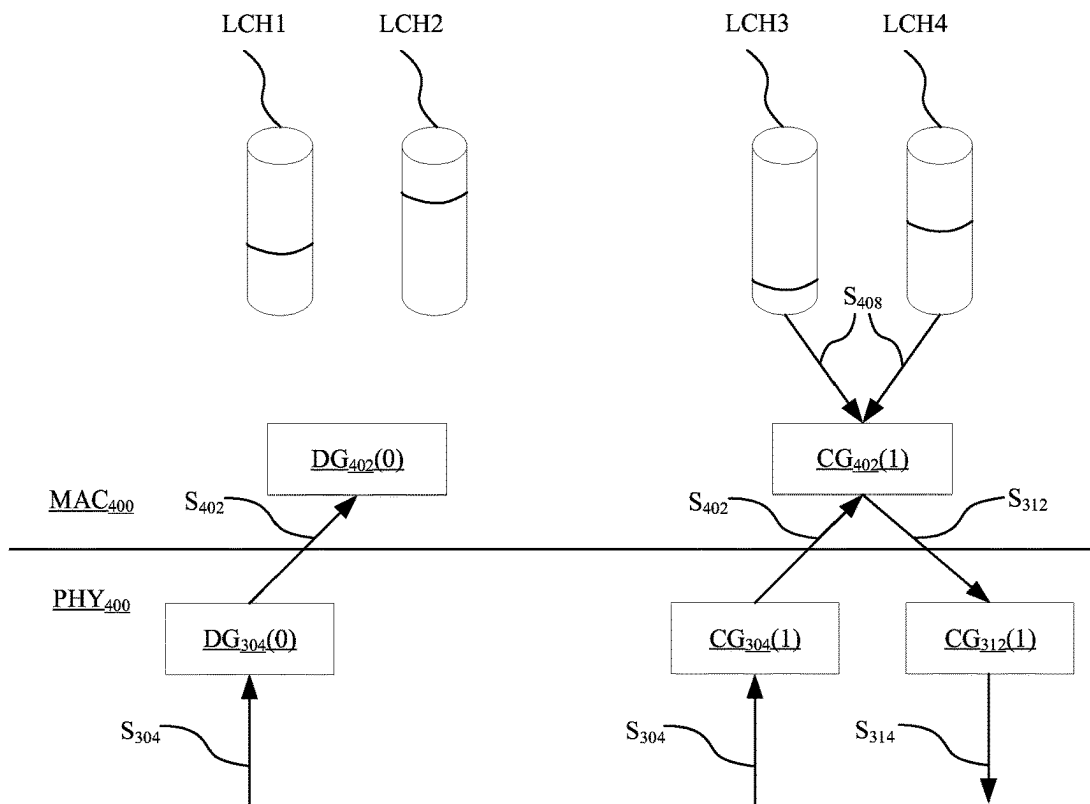


FIG. 6

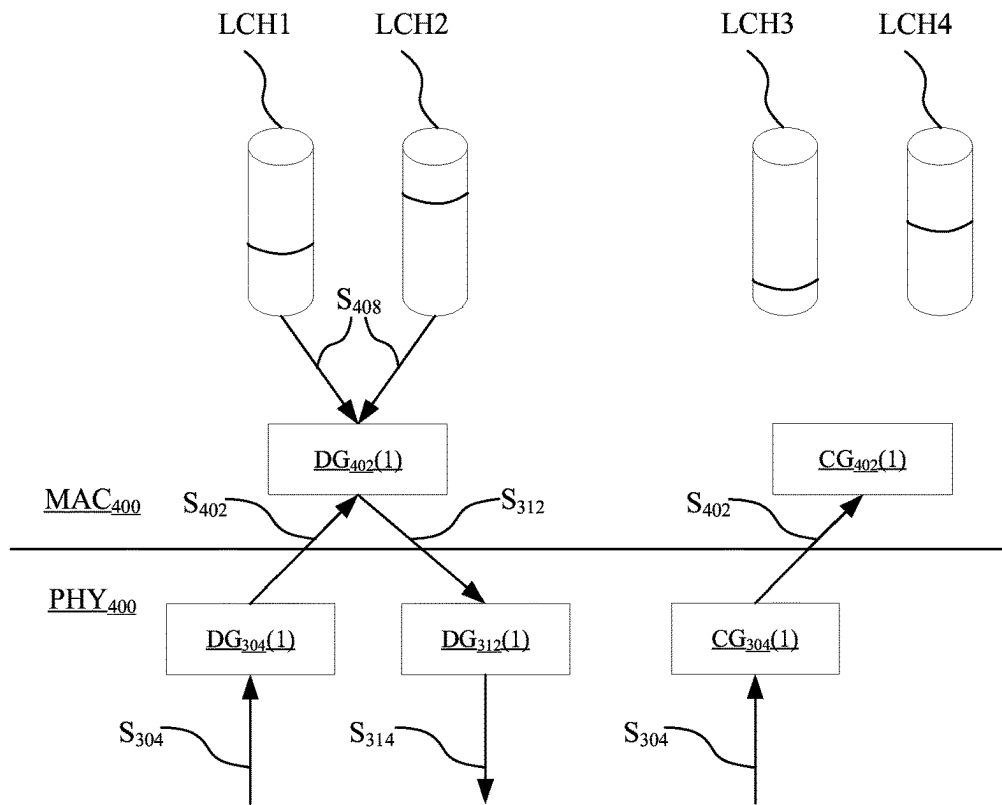


FIG. 7

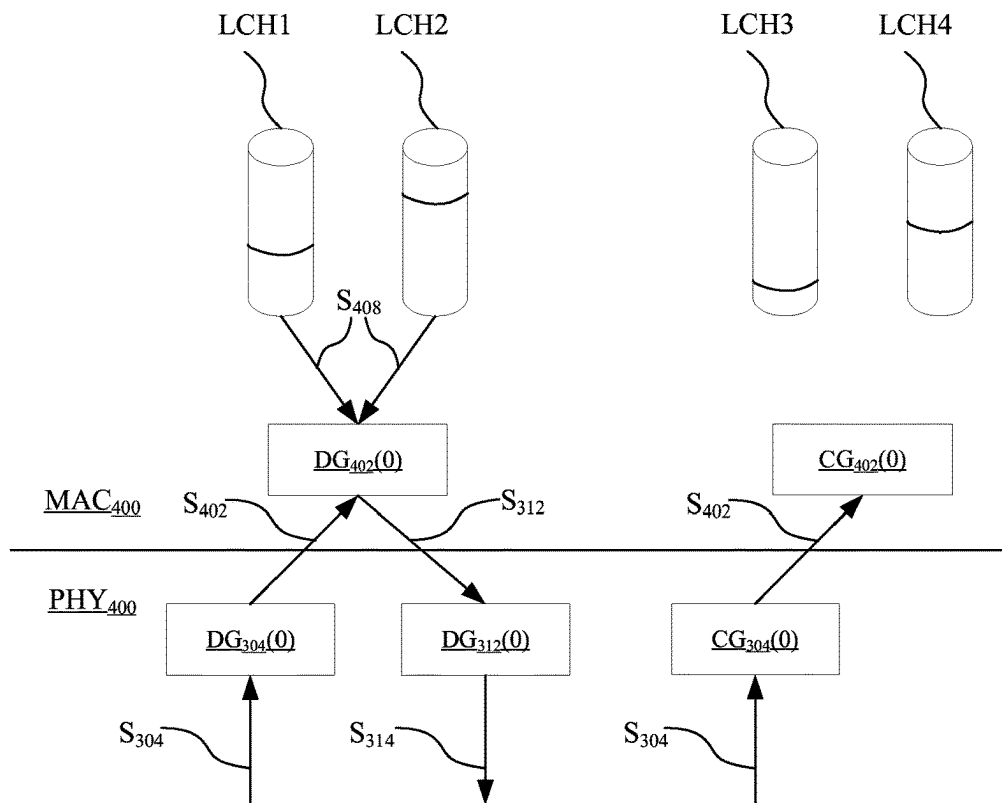


FIG. 8

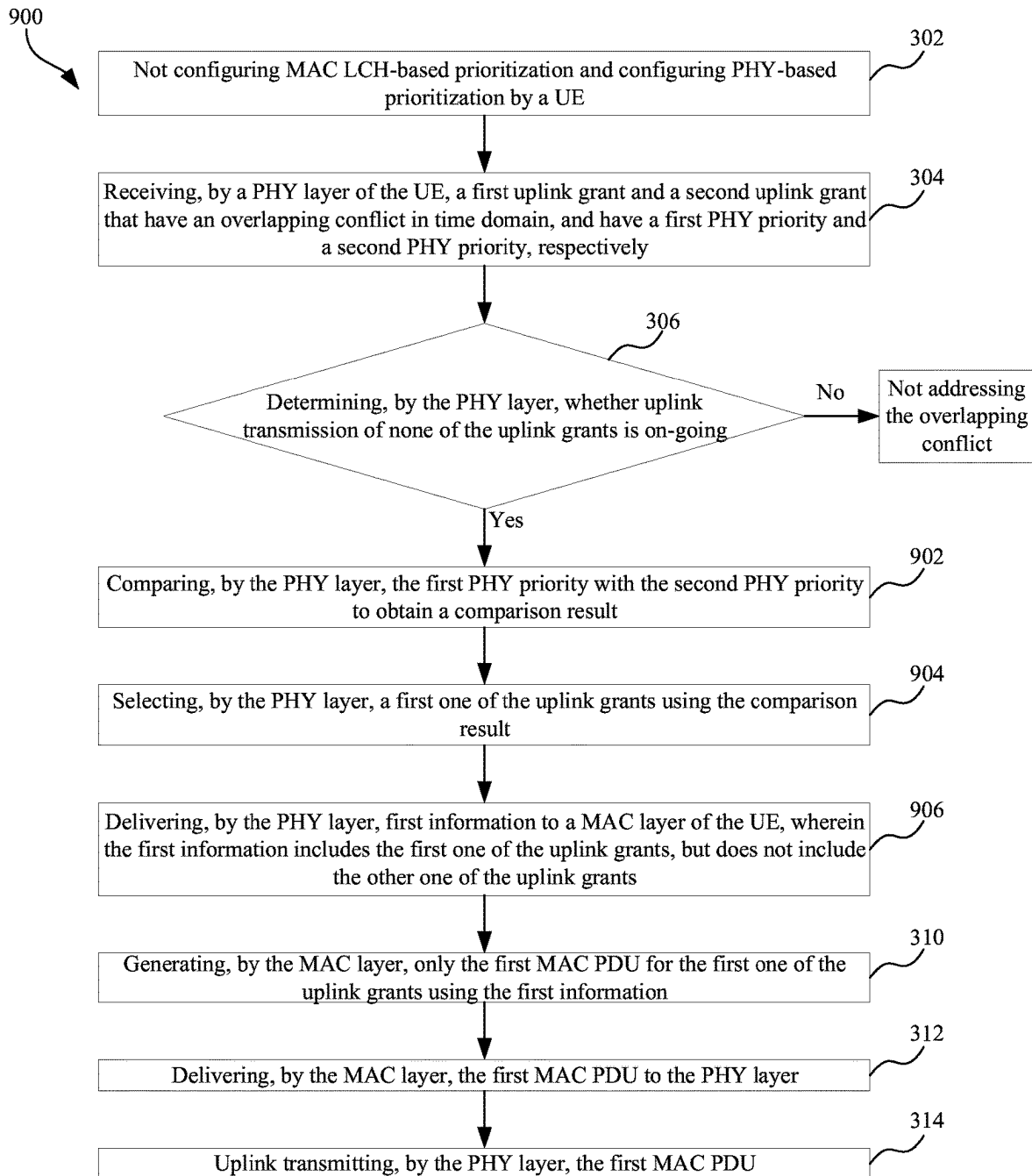


FIG. 9

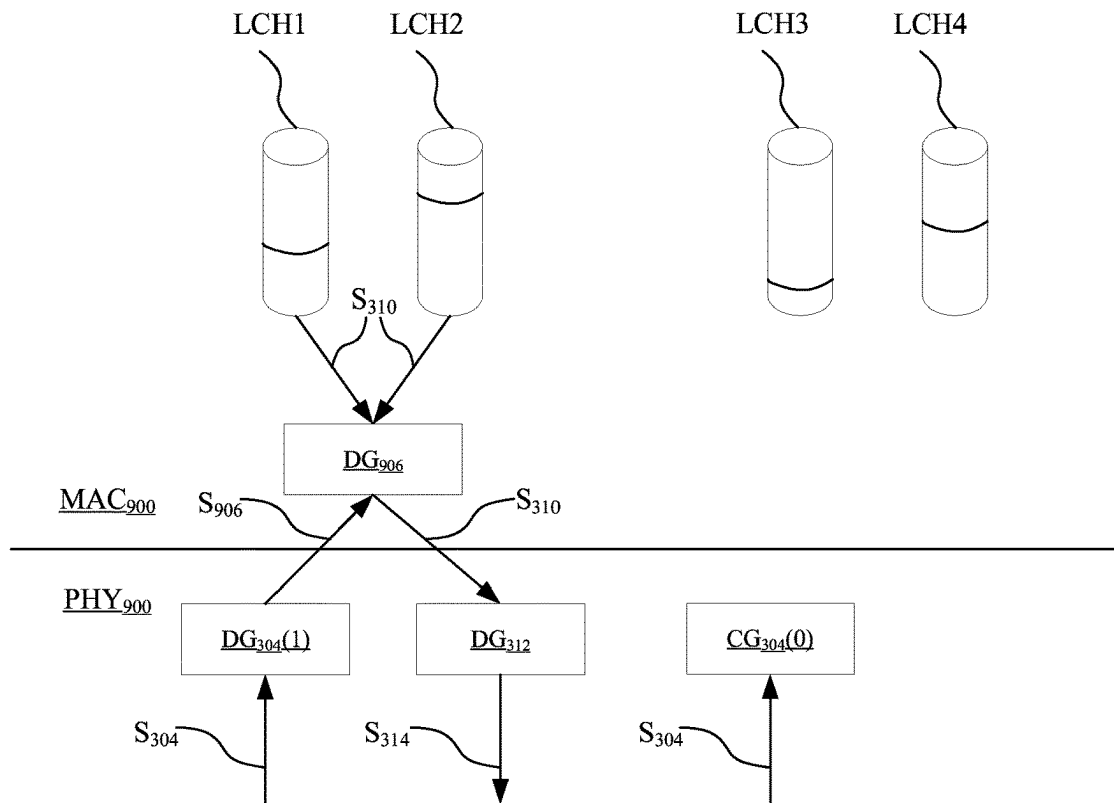


FIG. 10

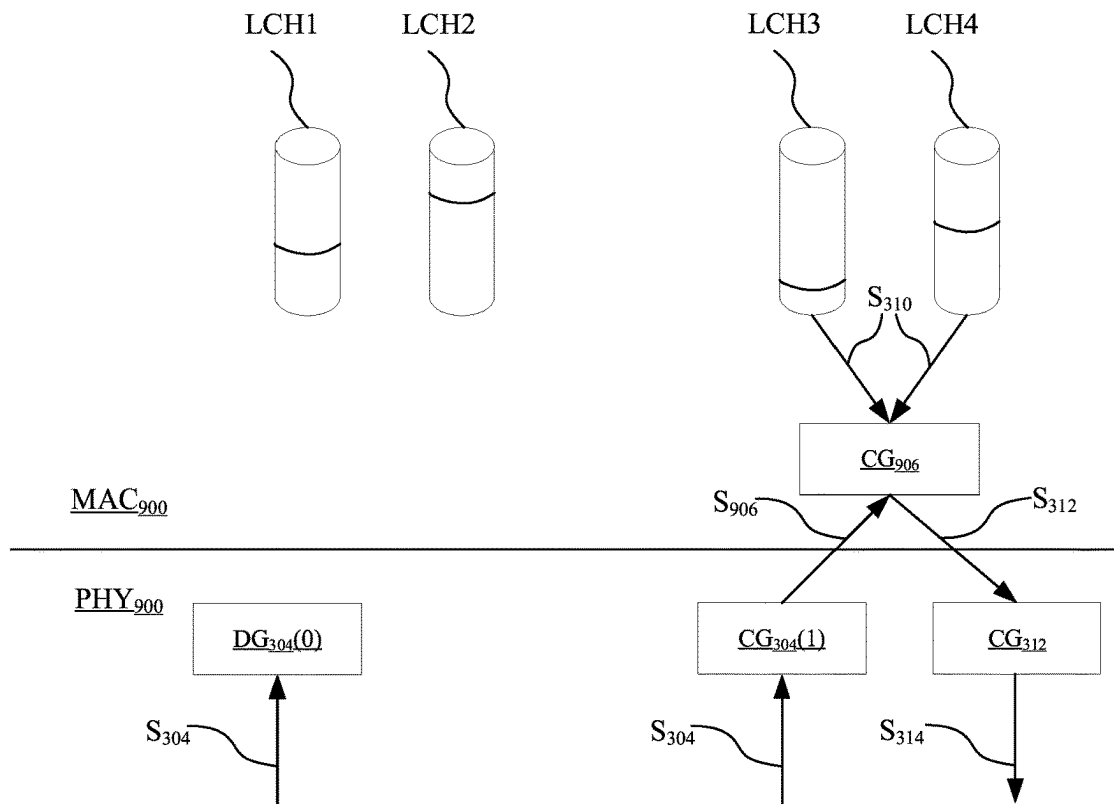


FIG. 11

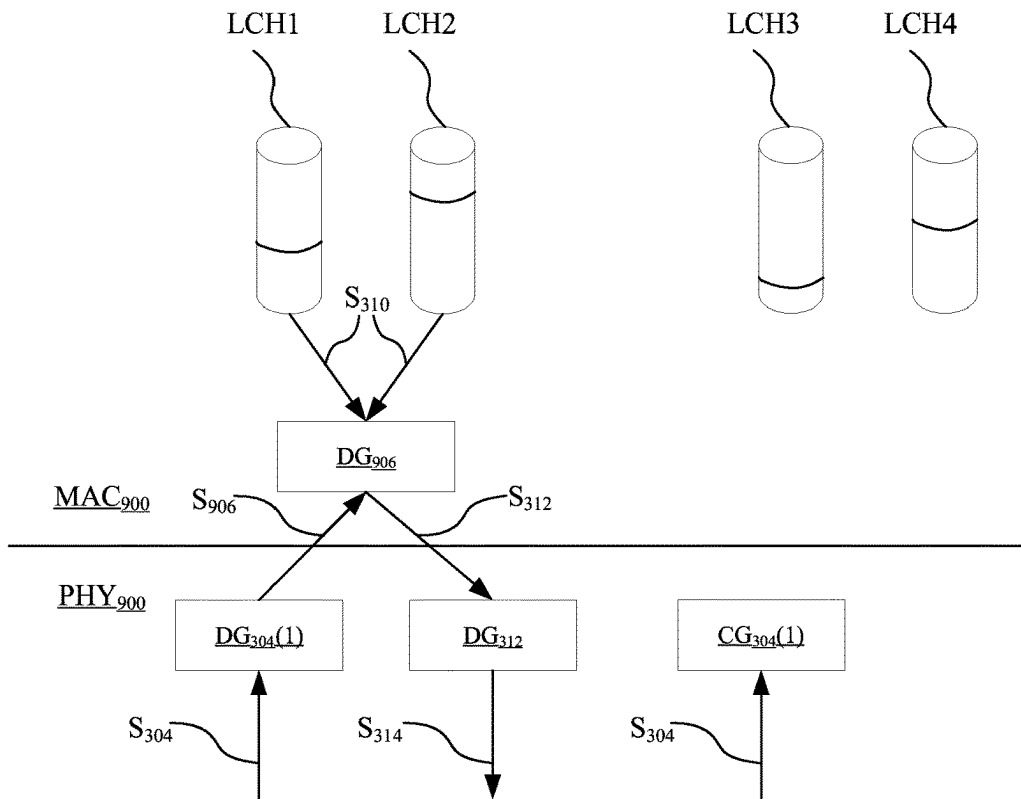


FIG. 12

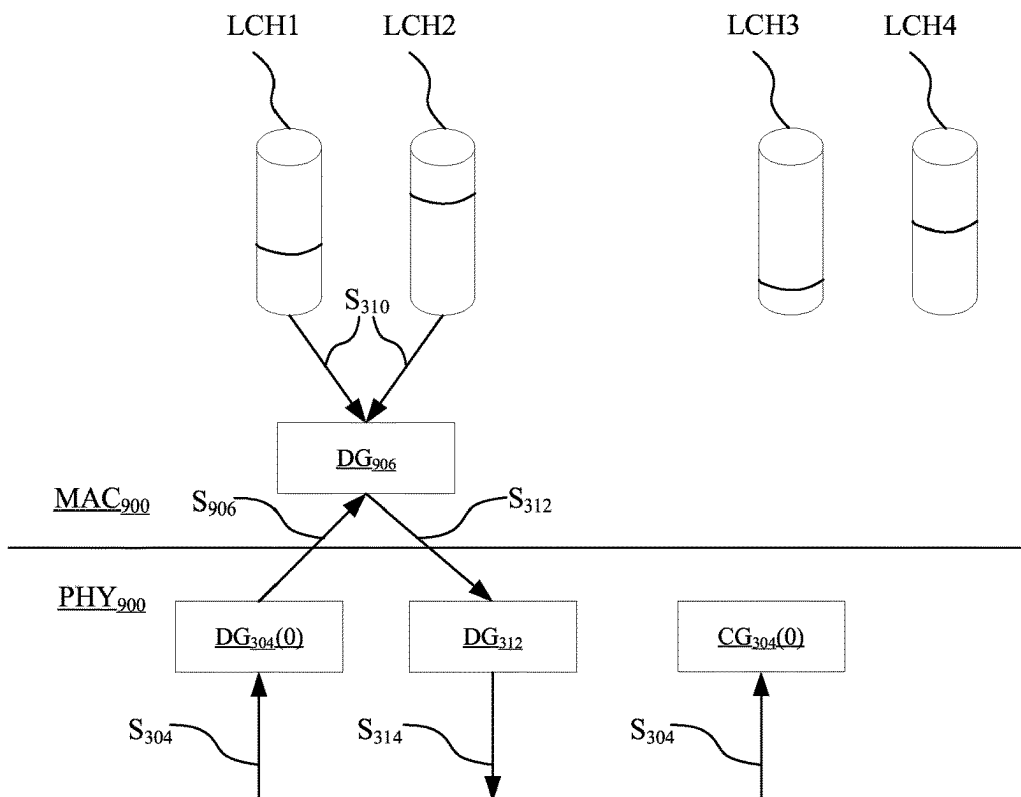


FIG. 13

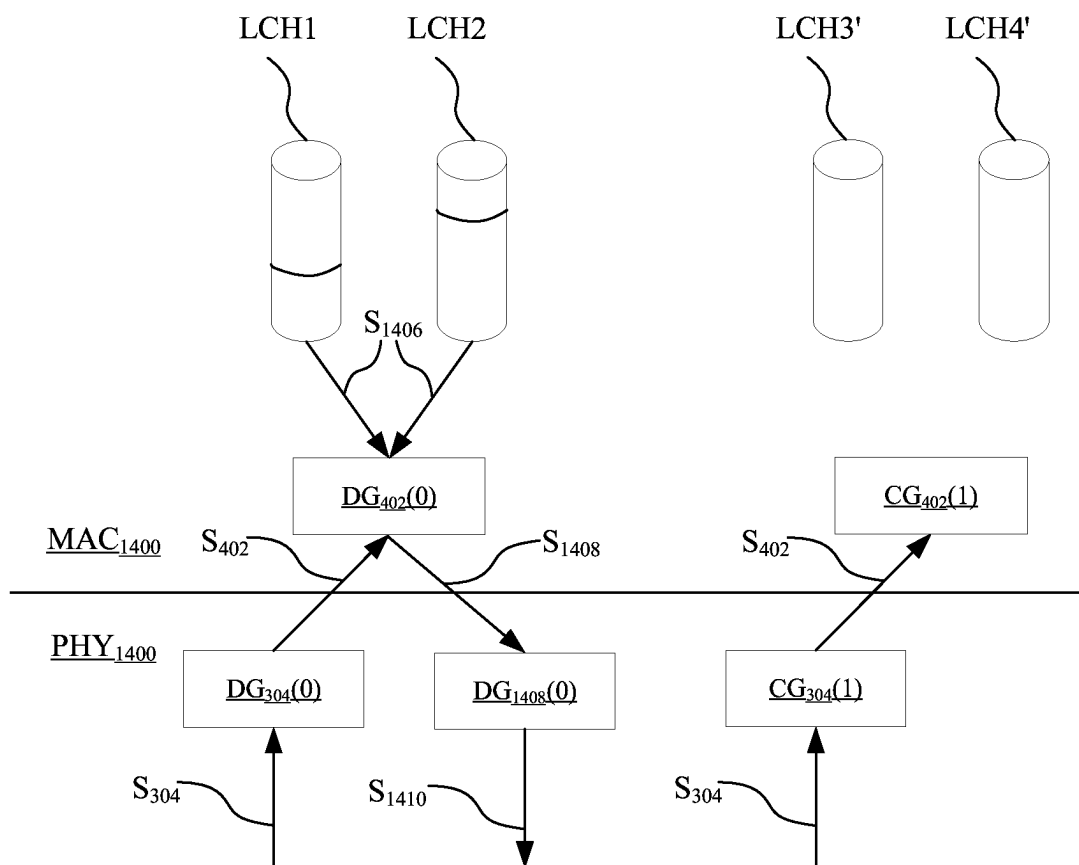


FIG. 14

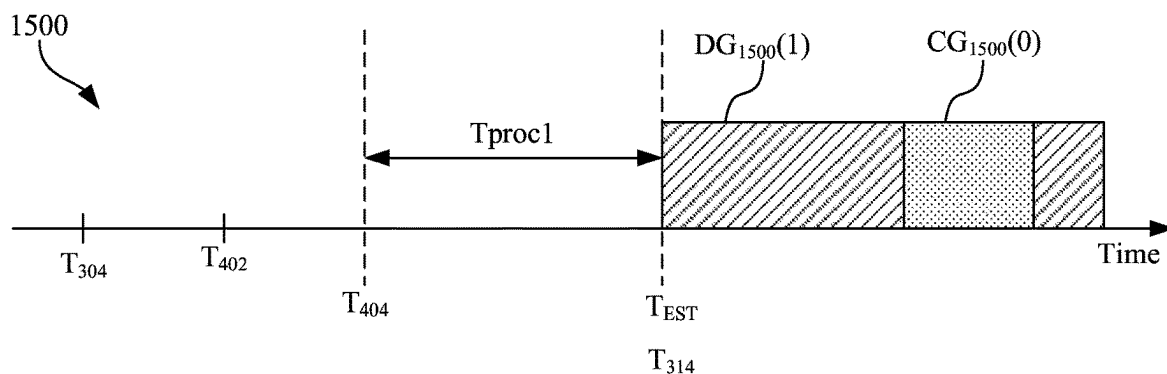


FIG. 15

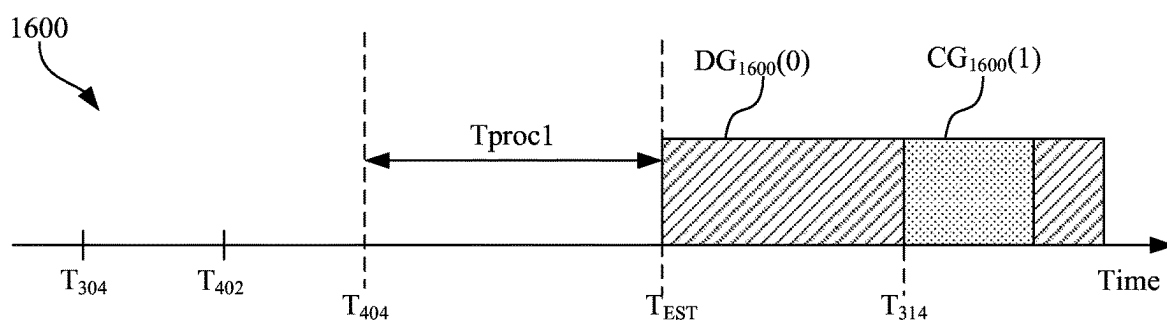


FIG. 16

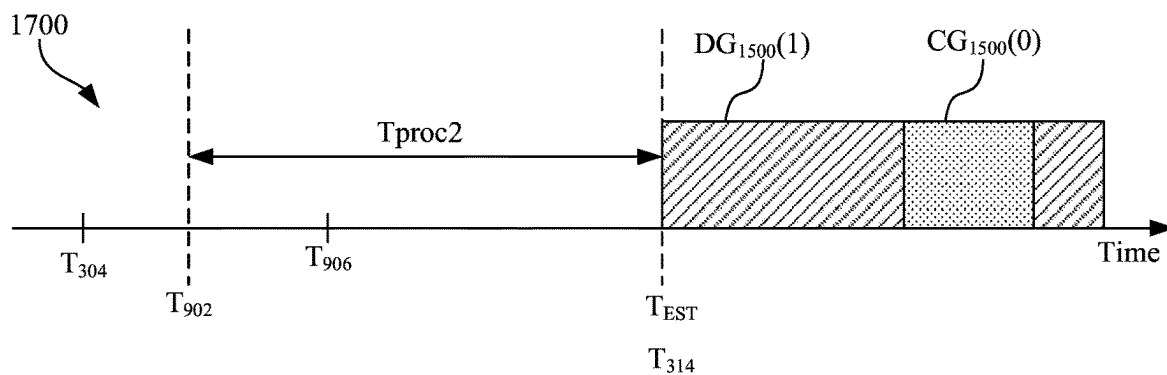


FIG. 17

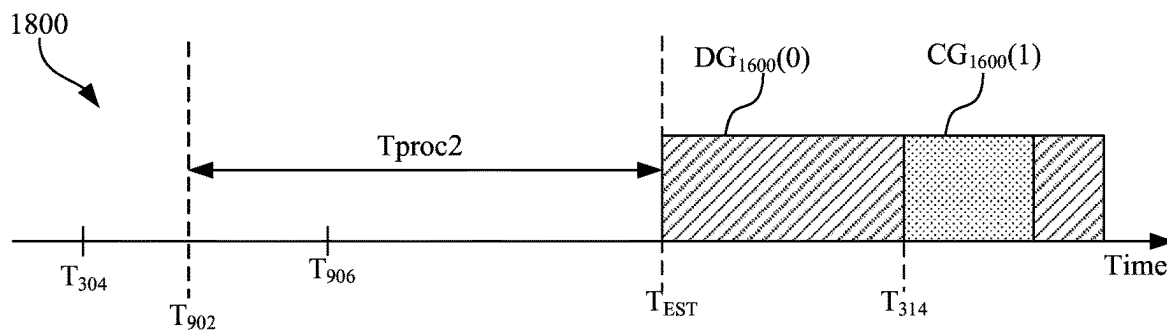


FIG. 18

1900



PHY priority	LCH priority
0000	Level-1
0001	Level-2
0010	Level-3
0011	Level-4
0100	Level-5
0101	Level-6
0110	Level-7
0111	Level-8
1000	Level-9
1001	Level-10
1010	Level-11
1011	Level-12
1100	Level-13
1101	Level-14
1110	Level-15
1111	Level-16

FIG. 19

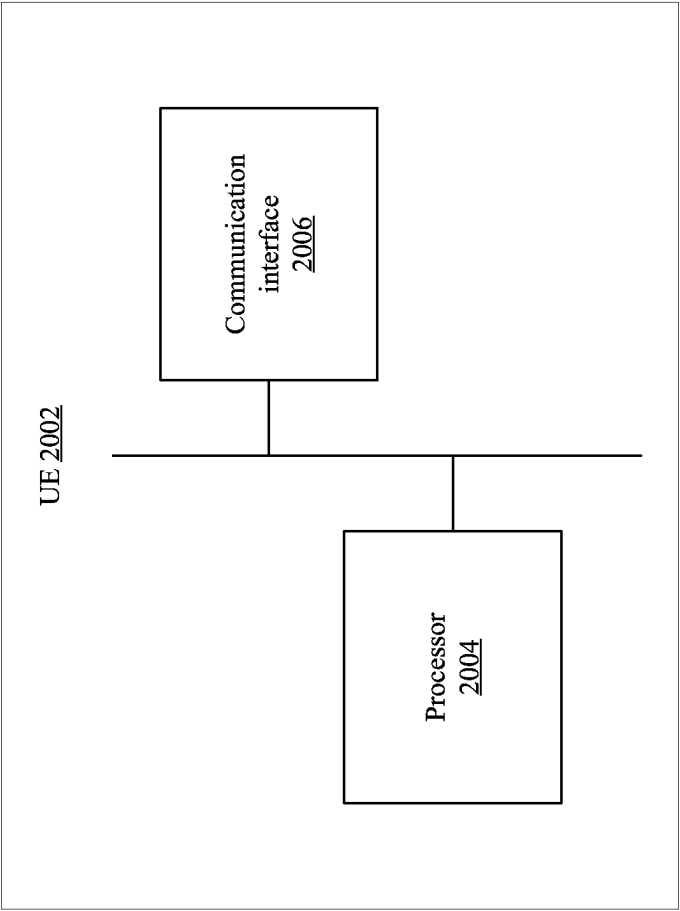
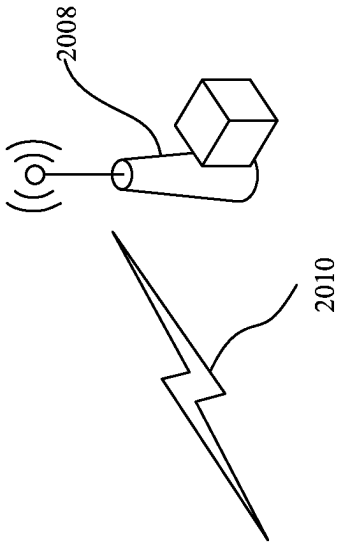


FIG. 20

**METHOD, USER EQUIPMENT, AND
NON-TRANSITORY COMPUTER READABLE
MEDIUM FOR PHYSICAL LAYER-BASED
PRIORITIZATION WITHOUT LOGICAL
CHANNEL-BASED PRIORITIZATION**

RELATED APPLICATIONS

This application is a National Phase of PCT Patent Application No. PCT/CN2021/111044 having International filing date of Aug. 5, 2021, which claims the benefit of priority under 35 USC § 119 (e) of U.S. Provisional Patent Application No. 63/062,179 filed on Aug. 6, 2020. The contents of the above applications are all incorporated by reference as if fully set forth herein in their entirety.

FIELD AND BACKGROUND OF THE
INVENTION

The present disclosure relates to the field of wireless communication, and more particularly, to a method, a user equipment, and computer readable-storage medium for physical layer-based prioritization without logical channel-based prioritization.

One type of an intra-user equipment (UE) uplink resource conflict is an overlapping conflict in time domain between two uplink grants. The two uplink grants can be a dynamic grant (DG) and a configured grant (CG), a CG and another CG, and a DG and another DG. The overlapping conflict in time domain may be resolved by physical layer-based prioritization (PHY-based prioritization) or media access control (MAC) logical channel-based prioritization (LCH-based prioritization).

For the PHY-based prioritization, a radio access network (RAN) node, such as a next generation NodeB (gNB), configures a physical layer priority (PHY priority) of a DG using a priority index in downlink control information (DCI). The RAN node configures a PHY priority of a CG using a priority index signaled by radio resource control (RRC) signaling. The PHY priority of each of the two uplink grants (the DG and the CG) is 2-level. According to the PHY-based prioritization rules, (1) in a case that a CG with a high PHY priority collides with a DG with a low PHY priority, the CG is prioritized, and if uplink transmission of the DG is on-going, the uplink transmission of the DG is canceled; (2) in a case that a CG with a high PHY priority collides with another CG with a low PHY priority, the CG with the high PHY priority is prioritized, the high-PHY priority CG may interrupt the on-going low-PHY priority CG, and a timeline therefor is up to UE implementation; (3) in a case that a CG with a low PHY priority collides with a DG with a high PHY priority, the DG is prioritized, and if uplink transmission of the CG is on-going, the uplink transmission of the CG is canceled (considering timeline restrictions); (4) in a case that a CG collides with a DG of the same PHY priority, the DG is prioritized, and since for the CG and the DG of the same PHY priority, a 3rd Generation Partnership Project Release 15 (3GPP Rel. 15) cancellation timeline applies, uplink transmission of the CG does not start; (5) a case that a CG collides with another CG of the same PHY priority is not addressed by the PHY-based prioritization; (6) a case that a DG collides with another DG is not addressed by the PHY-based prioritization.

For the LCH-based prioritization, a comparison on a priority (such as the highest priority) of all LCH(s) between the two uplink grants is performed to select one of the two uplink grants for which one MAC protocol data unit (PDU)

is generated. The LCH(s) are one or more LCHs that have data available and are multiplexed or can be multiplexed into each of the two uplink grants, taking into account logical channel prioritization (LCP) mapping restrictions.

The PHY-based prioritization and the LCH-based prioritization may complement each other in different situations.

SUMMARY OF THE INVENTION

For an existing method that uses physical layer-based prioritization (PHY-based prioritization) and media access control (MAC) logical channel-based prioritization (LCH-based prioritization) that complements each other in different situations to resolve an overlapping conflict in time domain between two uplink grants within a user equipment (UE), the UE needs to have both a capability of the PHY-based prioritization and a capability of the LCH-based prioritization. For industrial internet of things (IIoT) applications, it may be desirable that UEs are allowed to have versatile capabilities. Thus, the present disclosure proposes to enhance UE capability versatility by allowing a case that the PHY-based prioritization is configured and the LCH-based prioritization is not configured. Such configuration is also agreed upon in a RAN2#109bis-e meeting. When the existing method is modified in such a way that the LCH-based prioritization is not configured in situations that the LCH-based prioritization is originally used to resolve the overlapping conflict, two MAC protocol data units (PDUs) for the two uplink grants are generated, respectively, when uplink transmission for the uplink grants is not on-going, which can cause one of the MAC PDUs to be lost, one of the MAC PDUs needs to be retransmitted, or uplink transmission for one of the MAC PDUs needs to be cancelled. Thus, in the present disclosure, information for resolving the overlapping conflict by comparing the physical layer priorities (PHY priorities) of the two uplink grants, respectively, is delivered by the physical layer (PHY layer) to the MAC layer. In this way, only one MAC PDU is generated for one of the two uplink grants using the information.

In a first aspect of the present disclosure, a method performed by a UE includes: when MAC LCH-based prioritization is not configured, and PHY-based prioritization is configured in the UE, receiving, by a PHY layer of the UE, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively; and when uplink transmission for the uplink grants is not on-going, delivering, by the PHY layer, first information to a MAC layer of the UE, wherein the first information is for resolving the overlapping conflict by comparing the first PHY priority with the second PHY priority, and generating, by the MAC layer, only a first MAC PDU for a first one of the uplink grants using the first information.

In a second aspect of the present disclosure, a UE includes: at least one processor configured to execute instructions to implement the method of the first aspect of the present disclosure.

In a third aspect of the present disclosure, a non-transitory computer readable medium has instructions stored therein, wherein the method of the first aspect of the present disclosure is implemented when the instructions are executed.

BRIEF DESCRIPTION OF THE SEVERAL
VIEWS OF THE DRAWINGS

In order to more clearly illustrate the embodiments of the present disclosure or related art, the following figures will be

described in the embodiments are briefly introduced. It is obvious that the drawings are merely some embodiments of the present disclosure, a person having ordinary skill in this field can obtain other figures according to these figures without paying the premise.

FIG. 1 is a flowchart illustrating an existing method that is modified in such a way that media access control (MAC) logical channel-based prioritization (LCH-based prioritization) is not configured in situations that the MAC LCH-based prioritization is originally used to resolve an overlapping conflict.

FIG. 2 is a schematic diagram illustrating an example of a MAC protocol data unit (PDU) loss resulting from applying the modified existing method to an overlapping conflict in time domain between a dynamic grant (DG) with a high PHY priority and a configured grant (CG) with a low PHY priority.

FIG. 3 is a flowchart of a method performed by a user equipment (UE) for physical layer-based prioritization (PHY-based prioritization) without the LCH-based prioritization in accordance with some embodiments of the present disclosure.

FIG. 4 is a flowchart illustrating a method by a UE for PHY-based prioritization without MAC LCH-based prioritization, wherein PHY priority comparison for the PHY-based prioritization is performed by a MAC layer, in accordance with some embodiments of the present disclosure.

FIG. 5 is a schematic diagram illustrating an example of applying the method in FIG. 4 to an overlapping conflict in time domain between a DG with a high PHY priority and a CG with a low PHY priority.

FIG. 6 is a schematic diagram illustrating an example of applying the method in FIG. 4 to an overlapping conflict in time domain between a DG with a low PHY priority and a CG with a high PHY priority.

FIG. 7 is a schematic diagram illustrating an example of applying the method in FIG. 4 to an overlapping conflict in time domain between a DG with a high PHY priority and a CG with a high PHY priority.

FIG. 8 is a schematic diagram illustrating an example of applying the method in FIG. 4 to an overlapping conflict in time domain between a DG with a low PHY priority and a CG with a low PHY priority.

FIG. 9 is a flowchart illustrating a method by a UE for PHY-based prioritization without MAC LCH-based prioritization, wherein PHY priority comparison for the PHY-based prioritization is performed by a PHY layer, in accordance with some embodiments of the present disclosure.

FIG. 10 is a schematic diagram illustrating an example of applying the method in FIG. 9 to an overlapping conflict in time domain between a DG with a high PHY priority and a CG with a low PHY priority.

FIG. 11 is a schematic diagram illustrating an example of applying the method in FIG. 9 to an overlapping conflict in time domain between a DG with a low PHY priority and a CG with a high PHY priority.

FIG. 12 is a schematic diagram illustrating an example of applying the method in FIG. 9 to an overlapping conflict in time domain between a DG with a high PHY priority and a CG with a high PHY priority.

FIG. 13 is a schematic diagram illustrating an example of applying the method in FIG. 9 to an overlapping conflict in time domain between a DG with a low PHY priority and a CG with a low PHY priority.

FIG. 14 is a schematic diagram illustrating an example of applying the method in FIG. 4 that is modified so that comparison of PHY priorities in first information is per-

formed before determination of whether every LCH mapped to one of the uplink grants has no data.

FIG. 15 is a schematic timeline diagram illustrating the latest time for performing a step of comparing the first PHY priority (e.g., a high PHY priority of a DG) with the second PHY priority (e.g., a low PHY priority of a CG) by the MAC layer in the method in FIG. 4 in accordance with some embodiments of the present disclosure.

FIG. 16 is a schematic timeline diagram illustrating the latest time for performing a step of comparing the first PHY priority (e.g., a low PHY priority of a DG) with the second PHY priority (e.g., a high PHY priority of a CG) by the MAC layer in the method in FIG. 4 in accordance with some embodiments of the present disclosure.

FIG. 17 is a schematic timeline diagram illustrating the latest time for performing a step of comparing the first PHY priority (e.g., a high PHY priority of a DG) with the second PHY priority (e.g., a low PHY priority of a CG) by the PHY layer in the method in FIG. 9 in accordance with some embodiments of the present disclosure.

FIG. 18 is a schematic timeline diagram illustrating the latest time for performing a step of comparing the first PHY priority (e.g., a low PHY priority of a DG) with the second PHY priority (e.g., a high PHY priority of a CG) by the PHY layer in the method in FIG. 9 in accordance with some embodiments of the present disclosure.

FIG. 19 is a schematic table diagram illustrating a PHY priority-to-LCH priority mapping 1900, wherein the mapping is one-to-one, in accordance with some embodiments of the present disclosure.

FIG. 20 is a schematic diagram of a system including a UE that is capable of PHY-based prioritization without LCH-based prioritization and/or LCH-based prioritization without PHY-based prioritization in accordance with some embodiments of the present disclosure.

DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION

Embodiments of the present disclosure are described in detail with the technical matters, structural features, achieved objects, and effects with reference to the accompanying drawings as follows. Specifically, the terminologies in the embodiments of the present disclosure are merely for describing the purpose of the certain embodiment, and does not limit the disclosure.

As used here, when at least one operation is referred to as being performed “using”, “from”, “on”, or “on the basis of” at least one object, the at least one operation is performed “directly using”, “directly from”, “directly on”, or “directly on the basis of” the least one object, or at least one intervening operation can be present. In contrast, when the at least one operation is referred to as being performed “directly using”, “directly from”, “directly on”, or “directly on the basis of” the least one object, no intervening operation present.

A combination such as “at least one of A, B, or C,” “one or more of A, B, or C,” “at least one of A, B, and C,” “one or more of A, B, and C,” or “A, B, and/or C” may be A only, B only, C only, A and B, A and C, B and C, or A and B and C, where any combination may contain one or more members of A, B, or C.

For an existing method that uses physical layer-based prioritization (PHY-based prioritization) and media access control (MAC) logical channel-based prioritization (LCH-based prioritization) that complements each other in different situations to resolve an overlapping conflict in time

domain between two uplink grants within a user equipment (UE), the UE needs to have both a capability of the PHY-based prioritization and a capability of the LCH-based prioritization. For industrial internet of things (IIoT) applications, it may be desirable that UEs are allowed to have versatile capabilities. Thus, the present disclosure proposes to enhance UE capability versatility by allowing a case that the PHY-based prioritization is configured and the LCH-based prioritization is not configured. Such configuration is also agreed upon in a RAN2#109bis-e meeting. When the existing method is modified in such a way that the LCH-based prioritization is not configured in situations that the LCH-based prioritization is originally used to resolve the overlapping conflict, two MAC protocol data units (PDUs) for the two uplink grants are generated, respectively, when uplink transmission for the uplink grants is not on-going, which can cause one of the MAC PDUs to be lost, one of the MAC PDUs needs to be retransmitted, or uplink transmission for one of the MAC PDUs needs to be cancelled.

Referring to FIGS. 1 and 2, FIG. 1 is a flowchart illustrating the existing method 100 that is modified in such a way that the MAC LCH-based prioritization is not configured in situations that the MAC LCH-based prioritization is originally used to resolve the overlapping conflict. FIG. 2 is a schematic diagram illustrating an example of a MAC PDU loss resulting from applying the modified existing method to an overlapping conflict in time domain between a dynamic grant (DG) with a high PHY priority and a configured grant (CG) with a low PHY priority. The modified existing method 100 includes the following steps.

In step 102, MAC LCH-based prioritization is not configured and PHY-based prioritization is configured by a UE. When the LCH-based prioritization is not configured by the UE, the LCH-based prioritization is not performed. For the LCH-based prioritization, a comparison on a priority (such as the highest priority) of all logical channel(s) (LCH(s)) between the two uplink grants is performed to select one of the two uplink grants for which one MAC PDU is generated. The LCH(s) are one or more LCHs that have data available and are multiplexed or can be multiplexed into each of the two uplink grants, taking into account logical channel prioritization (LCP) mapping restrictions. When the PHY-based prioritization is configured by the UE, the PHY-based prioritization may be performed. For the PHY-based prioritization, a radio access network (RAN) node, such as a next generation NodeB (gNB), configures a physical layer priority (PHY priority) of a DG using a priority index in downlink control information (DCI). The RAN node configures a PHY priority of a CG using a priority index signaled by radio resource control (RRC) signaling. The PHY priority of each of the two uplink grants (the DG and the CG) is 2-level. The PHY priority is high when the priority index is 1 and is low when the priority index is 0. According to the PHY-based prioritization rules, (1) in a case that a CG with a high PHY priority collides with a DG with a low PHY priority, the CG is prioritized, and if uplink transmission of the DG is on-going, the uplink transmission of the DG is canceled; (2) in a case that a CG with a high PHY priority collides with another CG with a low PHY priority, the CG with the high PHY priority is prioritized, the high-PHY priority CG may interrupt the on-going low-PHY priority CG, and a timeline therefor is up to UE implementation; (3) in a case that a CG with a low PHY priority collides with a DG with a high PHY priority, the DG is prioritized, and if uplink transmission of the CG is on-going, the uplink transmission of the CG is canceled (considering timeline restrictions); (4) in a case that a CG collides with a DG of the same PHY priority, the

DG is prioritized, and since for the CG and the DG of the same PHY priority, a 3rd Generation Partnership Project Release 15 (3GPP Rel. 15) cancellation timeline applies, uplink transmission of the CG does not start; (5) a case that a CG collides with another CG of the same PHY priority is not addressed by the PHY-based prioritization; (6) a case that a DG collides with another DG is not addressed by the PHY-based prioritization.

In step 104, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively, is received by a PHY layer of the UE. Illustratively, in the example in FIG. 2, the first uplink grant is a DG with a high PHY priority $DG_{104}(1)$ (i.e., with a priority index of 1 for the first PHY priority). The second uplink grant is a CG with a low PHY priority $CG_{104}(0)$ (i.e., with a priority index of 0 for the second PHY priority). The DG $DG_{104}(1)$ and the CG $CG_{104}(0)$ have an overlapping conflict in time domain, i.e., a physical uplink shared channel (PUSCH) duration of the DG $DG_{104}(1)$ overlaps a PUSCH duration of the CG $CG_{104}(0)$. In step 104, a PHY layer PHY_{100} of the UE performs a step S_{104} of receiving the DG $DG_{104}(1)$ and the CG $CG_{104}(0)$.

In step 106, the uplink grants are delivered by the PHY layer to a MAC layer of the UE. In the example in FIG. 2, in step 106, the PHY layer PHY_{100} performs a step S_{106} of delivering the DG without the first PHY priority DG_{106} and the CG without the second PHY priority CG_{106} to a MAC layer MAC_{100} of the UE. The DG DG_{106} has the same content as the DG $DG_{104}(1)$ except the DG DG_{106} does not have information of the first PHY priority. The CG CG_{106} has the same content as the CG $CG_{104}(0)$ except the CG CG_{106} does not have information of the second PHY priority.

In step 108, when uplink transmission for the uplink grants is not on-going, two MAC PDUs for the uplink grants, respectively, are generated by the MAC layer. In the example in FIG. 2, in step 108, when uplink transmission for the uplink grants (the DG DG_{106} and the CG CG_{106}) is not on-going, the MAC layer MAC_{100} performs a step S_{108} of encapsulating, on the basis of LCP mapping restrictions, data from an LCH $LCH1$ and an LCH $LCH2$ into a first MAC PDU for the DG DG_{106} and data from an LCH $LCH3$ and an LCH $LCH4$ into a second MAC PDU for the CG CG_{106} . Thus, because the LCH-based prioritization is not configured in step 102, two MAC PDUs (i.e., the first MAC PDU and the second MAC PDU) for the uplink grants (i.e., the DG DG_{106} and the CG CG_{106}), respectively, are generated by the MAC layer MAC_{100} . The LCP mapping restrictions for each LCH $LCH1$, . . . , or $LCH4$ is configured using parameters including allowedSCS-List, maxPUSCH-Duration, configuredGrantType1Allowed, and allowedServingCells. For example, configuredGrantType1Allowed of each of the LCHs $LCH3$ and $LCH4$ is enabled. Thus, the data from $LCH3$ and $LCH4$ can be encapsulated into the second MAC PDU for the CG CG_{106} . The data from $LCH3$ and $LCH4$ may alternatively be encapsulated into the first MAC PDU for the DG DG_{106} when maxPUSCH-Duration is satisfied. On the other hand, configuredGrantType1Allowed of each of the LCHs $LCH1$ and $LCH2$ is disabled. Thus, the data from $LCH1$ and $LCH2$ can only be encapsulated into the first MAC PDU for the DG DG_{106} .

In step 110, the two MAC PDUs are delivered by the MAC layer to the PHY layer. In the example in FIG. 2, in step 110, the MAC layer MAC_{100} performs a step S_{110} of delivering the first MAC PDU for the DG DG_{106} and the

second MAC PDU for the CG CG₁₀₆ to the PHY layer PHY₁₀₀. The DG DG₁₀₆ of which the first MAC PDU is received by the PHY layer PHY₁₀₀ along with the corresponding PHY priority of the DG DG₁₀₆ known by the PHY layer PHY₁₀₀ is denoted as DG₁₁₀(1). The CG CG₁₀₆ of which the second MAC PDU is received by the PHY layer PHY₁₀₀ along with the corresponding PHY priority of the CG CG₁₀₆ known by the PHY layer PHY₁₀₀ is denoted as CG₁₁₀(0).

In step 112, one of the two MAC PDUs is uplink transmitted by the PHY layer according to a rule of the PHY-based prioritization. In the example in FIG. 2, because the first MAC PDU is for the DG DG₁₁₀(1) and the second MAC PDU is for the CG CG₁₁₀(0), rule (3) of the PHY-based prioritization rules described for step 102 applies. According to rule (3) of the PHY-based prioritization rules, the DG DG₁₁₀(1) is prioritized. Thus, the PHY layer PHY₁₀₀ performs a step S₁₁₂ of uplink transmitting the first MAC PDU for the DG DG₁₁₀(1) and performs a step S₁₁₄ of not uplink transmitting the second MAC PDU for the CG CG₁₁₀(0). When the second MAC PDU for the CG CG₁₁₀(0) is not uplink transmitted to the RAN node, because the RAN node does not know whether the second MAC PDU is not transmitted, or there is no second MAC PDU because every LCH mapped to the CG CG₁₁₀(0) has no data, the RAN node may not issue a retransmission grant. Thus, the second MAC PDU is lost. Alternatively, uplink transmission for the second MAC PDU may be on-going when uplink transmission for the first MAC PDU starts, and the uplink transmission for the second MAC PDU is cancelled. The generation and cancellation of the second MAC PDU may be inefficient because uplink transmission for the two uplink grants (the DG DG₁₁₀(1) and the CG CG₁₁₀(0)) is not on-going when the first MAC PDU and the second MAC PDU are generated and the first PHY priority of the DG DG₁₀₄(1) and the second PHY priority of the CG CG₁₀₄(0) are known when the DG DG₁₀₄(1) and the CG CG₁₀₄(0) are received in step 104.

For other levels of the PHY priorities of the DG and the CG, it can be understood on the basis of the above example that the modified existing method 100 has a deficiency of a MAC PDU loss, inefficiency due to a need for retransmission, or inefficiency due to a need for cancellation.

Thus, in the present disclosure, information for resolving the overlapping conflict by comparing the PHY priorities of the two uplink grants, respectively, is delivered by the PHY layer to the MAC layer. In this way, only one MAC PDU is generated for one of the two uplink grants using the information.

Referring to FIG. 3, FIG. 3 is a flowchart of a method 300 performed by a UE for PHY-based prioritization without MAC LCH-based prioritization in accordance with some embodiments of the present disclosure. The method 300 includes the following steps.

In step 302, MAC LCH-based prioritization is not configured and PHY-based prioritization is configured by the UE. As used here, when a capability of a UE such as MAC LCH-based prioritization or PHY-based prioritization of the UE is referred to as being “configured” or “not configured”, the UE may support the capability and the capability may be “enabled” (corresponding to “configured”) or “disabled” (corresponding to “not configured”), or the UE may “support” (corresponding to “configured”) or “not support” (corresponding to “not configured”) the capability. When the LCH-based prioritization is not configured by the UE, the LCH-based prioritization is not performed. When the PHY-based prioritization is configured by the UE, the PHY-based

prioritization may be performed. The LCH-based prioritization has been described with reference to step 102 in FIG. 1 and is omitted here. An embodiment of rules of the PHY-based prioritization has been described with reference to step 102 in FIG. 1 and is omitted here. Other rules of the PHY-based prioritization that are known in the art to replace the corresponding rules of the PHY-based prioritization and be suitable for resolving an overlapping conflict by comparing PHY priorities of two uplink grants are within the contemplated scope of the present disclosure.

In step 304, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively, is received by a PHY layer of the UE. Embodiments of step 304 illustrated with examples are to be described with reference to step 304 in FIG. 4 and step 304 in FIG. 9.

In step 306, whether uplink transmission for the uplink grants is not on-going is determined by the PHY layer. If so, the PHY-based prioritization of the method 300 is performed and thus, the steps 308, 310, 312, and 314 are performed. If not, the overlapping conflict is not addressed by the PHY-based prioritization of the method 300 because it is possible that cancellation needs to be performed.

In step 308, first information is delivered by the PHY layer to a MAC layer of the UE, wherein the first information is for resolving the overlapping conflict by comparing the first PHY priority with the second PHY priority. As used here, when information is referred to as being “for resolving the overlapping conflict by comparing the first PHY priority with the second PHY priority”, the information explicitly or implicitly indicates the first PHY priority and the second PHY priority (i.e., possible levels of the PHY priorities can be directly or indirectly known from the information). For the information that explicitly indicates the first PHY priority and the second PHY priority, a comparison can be made between the first PHY priority and the second PHY priority in the information to resolve the overlapping conflict (i.e., select one of the uplink grants in the information) according to a rule of the PHY-based prioritization mentioned in step 302. For the information that implicitly indicates the first PHY priority and the second PHY priority, a comparison has been made between the first PHY priority and the second PHY priority to resolve the overlapping conflict (i.e., select one of the uplink grants) according to a rule of the PHY-based prioritization mentioned in step 302, and the information includes the selected one of the uplink grants but not the other one of the grants. Embodiments for explicit indication illustrated with examples are to be described with reference to step 402 and corresponding steps 404, 406, and 408 in FIG. 4. Embodiments for implicit indication illustrated with examples are to be described with reference to step 906, and corresponding steps 902 and 904 in FIG. 9.

In step 310, only a first MAC PDU for a first one of the uplink grants is generated using the first information by the MAC layer. Embodiments of step 310 illustrated with examples are to be described with reference to step 408 in FIG. 4 and step 310 in FIG. 9.

In step 312, the first MAC PDU is delivered by the MAC layer to the PHY layer. Embodiments of step 312 illustrated with examples are to be described with reference to step 312 in FIG. 4 and step 312 in FIG. 9.

In step 314, the first MAC PDU is uplink transmitted by the PHY layer. Embodiments of step 314 illustrated with examples are to be described with reference to step 314 in FIG. 4 and step 314 in FIG. 9.

Because the first information for resolving the overlapping conflict by comparing the first PHY priority with the second PHY priority is used for MAC PDU generation, only the first MAC PDU for the first one of the uplink grants is generated compared with the modified existing method 100 that generates two MAC PDUs generated in step 108. Thus, the deficiency of the MAC PDU loss, the inefficiency due to the need for retransmission, or the inefficiency due to the need for cancellation for the modified existing method 100 is improved.

Referring to FIGS. 4 and 5, FIG. 4 is a flowchart illustrating a method 400 by a UE for PHY-based prioritization without MAC LCH-based prioritization, wherein PHY priority comparison for the PHY-based prioritization is performed by a MAC layer, in accordance with some embodiments of the present disclosure. FIG. 5 is a schematic diagram illustrating an example of applying the method 400 to an overlapping conflict in time domain between a DG with a high PHY priority and a CG with a low PHY priority. The method 400 includes the following steps.

In step 302, MAC LCH-based prioritization is not configured and PHY-based prioritization is configured by a UE. The step 302 has been described with reference to step 302 in FIG. 3 and is omitted here.

In step 304, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively, is received by a PHY layer of the UE. Illustratively, in the example in FIG. 5, the first uplink grant is a DG with a high PHY priority $DG_{304}(1)$ (i.e., with a priority index of 1 for the first PHY priority). The second uplink grant is a CG with a low PHY priority $CG_{304}(0)$ (i.e., with a priority index of 0 for the second PHY priority). The $DG_{304}(1)$ and the $CG_{304}(0)$ have an overlapping conflict in time domain, i.e., a PUSCH duration of the $DG_{304}(1)$ overlaps a PUSCH duration of the $CG_{304}(0)$. In step 304, a PHY layer PHY_{400} of the UE performs a step S_{304} of receiving the $DG_{304}(1)$ and the $CG_{304}(0)$.

In step 306, whether uplink transmission for the uplink grants is not on-going is determined by the PHY layer. The step 306 has been described with reference to step 306 in FIG. 3 and is omitted here.

In step 402, first information is delivered by the PHY layer to a MAC layer of the UE, wherein the first information includes the first uplink grant and the second uplink grant, and the first PHY priority and the second PHY priority. Step 402 is an embodiment of the step 308. In this embodiment, the first information explicitly indicates the first PHY priority and the second PHY priority. In the example in FIG. 5, in step 402, the PHY layer PHY_{400} performs a step S_{402} of delivering first information that includes the DG with the high PHY priority $DG_{304}(1)$ and the CG with the low PHY priority $CG_{304}(0)$ to a MAC layer MAC_{400} of the UE. The DG with the high PHY priority in the MAC layer MAC_{400} has the same content as the DG with the high PHY priority $DG_{304}(1)$ in the PHY layer PHY_{400} and is denoted as $DG_{402}(1)$. The CG with the low PHY priority in the MAC layer MAC_{400} has the same content as the CG with the low PHY priority $CG_{304}(0)$ is denoted as $CG_{402}(0)$.

The step 310 in FIG. 3 includes the steps 404, 406, and 408 in FIG. 4.

In step 404, the first PHY priority is compared with the second PHY priority to obtain a comparison result by the MAC layer. In the example in FIG. 5, in step 404, the MAC layer MAC_{400} compares the first PHY priority of the DG with the high PHY priority $DG_{402}(1)$ and the second PHY

priority of the CG with the low PHY priority $CG_{402}(0)$ to obtain that the first PHY priority of the DG $DG_{402}(1)$ is higher than the second PHY priority of the CG $CG_{402}(0)$ as a comparison result.

In step 406, the first one of the uplink grants is selected using the comparison result by the MAC layer. In the example in FIG. 5, in step 406, because the first PHY priority of the DG $DG_{402}(1)$ is higher than the second PHY priority of the CG $CG_{402}(0)$ in the comparison result, according to rule (3) of the PHY-based prioritization in step 302 described with reference to FIG. 3, the MAC layer MAC_{400} selects the DG $DG_{402}(1)$ (i.e., the DG $DG_{402}(1)$ is the first one of the uplink grants).

In step 408, only the first MAC PDU for the first one of the uplink grants is generated by the MAC layer. In the example in FIG. 5, in step 408, the MAC layer MAC_{400} performs a step S_{408} of encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH1 and an LCH LCH2 into a first MAC PDU for the DG $DG_{402}(1)$ and not encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH3 and an LCH LCH4 into a second MAC PDU for the CG $CG_{402}(0)$. Thus, even when the LCH-based prioritization is not configured in step 302, only the first MAC PDU for the DG $DG_{402}(1)$ is generated. Examples of the LCP mapping restrictions have been described with reference to step 108 in FIG. 1 and are omitted here.

In step 312, the first MAC PDU is delivered by the MAC layer to the PHY layer. In the example in FIG. 5, in step 312, the MAC layer MAC_{400} performs a step S_{312} of delivering the first MAC PDU for the DG $DG_{402}(1)$ to the PHY layer PHY_{400} . The DG $DG_{402}(1)$ of which the first MAC PDU is received by the PHY layer PHY_{400} is denoted as $DG_{312}(1)$. Because the second MAC PDU for the CG $CG_{402}(0)$ is not generated, the second MAC PDU for the CG $CG_{402}(0)$ is not delivered by the MAC layer MAC_{400} to the PHY layer PHY_{400} compared with the modified existing method 100 that delivers the second MAC PDU for the CG CG_{106} by the MAC layer MAC_{100} to the PHY layer PHY_{100} in step 110 which results in the second MAC PDU for the CG $CG_{110}(0)$ in the PHY layer PHY_{100} .

In step 314, the first MAC PDU is uplink transmitted by the PHY layer. In the example in FIG. 5, the PHY layer PHY_{400} performs a step S_{314} of uplink transmitting the first MAC PDU for the DG $DG_{312}(1)$. Because the second MAC PDU for the CG $CG_{402}(0)$ is not generated, the deficiency of the loss of the second MAC PDU or the inefficiency due to the need for cancellation of the second MAC PDU is improved.

Referring to FIGS. 4 and 6, FIG. 4 is a flowchart illustrating a method 400 by a UE for PHY-based prioritization without MAC LCH-based prioritization, wherein PHY priority comparison for the PHY-based prioritization is performed by a MAC layer, in accordance with some embodiments of the present disclosure. FIG. 6 is a schematic diagram illustrating an example of applying the method 400 to an overlapping conflict in time domain between a DG with a low PHY priority and a CG with a high PHY priority. The method 400 includes the following steps.

In step 302, MAC LCH-based prioritization is not configured and PHY-based prioritization is configured by a UE. The step 302 has been described with reference to step 302 in FIG. 3 and is omitted here.

In step 304, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively, is received by a PHY layer of the UE. Illustratively, in the

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example in FIG. 6, the first uplink grant is a DG with a low PHY priority $DG_{304}(0)$ (i.e., with a priority index of 0 for the first PHY priority). The second uplink grant is a CG with a high PHY priority $CG_{304}(1)$ (i.e., with a priority index of 1 for the second PHY priority). The DG $DG_{304}(0)$ and the CG $CG_{304}(1)$ have an overlapping conflict in time domain, i.e., a PUSCH duration of the DG $DG_{304}(0)$ overlaps a PUSCH duration of the CG $CG_{304}(1)$. In step 304, a PHY layer PHY_{400} of the UE performs a step S_{304} of receiving the DG $DG_{304}(0)$ and the CG $CG_{304}(1)$.

In step 306, whether uplink transmission for the uplink grants is not on-going is determined by the PHY layer. The step 306 has been described with reference to step 306 in FIG. 3 and is omitted here.

In step 402, first information is delivered by the PHY layer to a MAC layer of the UE, wherein the first information includes the first uplink grant and the second uplink grant, and the first PHY priority and the second PHY priority. Step 402 is an embodiment of the step 308. In this embodiment, the first information explicitly indicates the first PHY priority and the second PHY priority. In the example in FIG. 6, in step 402, the PHY layer PHY_{400} performs a step S_{402} of delivering first information that includes the DG with the low PHY priority $DG_{304}(0)$ and the CG with the high PHY priority $CG_{304}(1)$ to a MAC layer MAC_{400} of the UE. The DG with the low PHY priority in the MAC layer MAC_{400} has the same content as the DG with the low PHY priority $DG_{304}(0)$ in the PHY layer PHY_{400} and is denoted as $DG_{402}(0)$. The CG with the high PHY priority in the MAC layer MAC_{400} has the same content as the CG with the high PHY priority $CG_{304}(1)$ is denoted as $CG_{402}(1)$.

The step 310 in FIG. 3 includes the steps 404, 406, and 408 in FIG. 4.

In step 404, the first PHY priority is compared with the second PHY priority to obtain a comparison result by the MAC layer. In the example in FIG. 6, in step 404, the MAC layer MAC_{400} compares the first PHY priority of the DG with the low PHY priority $DG_{402}(0)$ and the second PHY priority of the CG with the high PHY priority $CG_{402}(1)$ to obtain that the second PHY priority of the CG $CG_{402}(1)$ is higher than the first PHY priority of the DG $DG_{402}(0)$ as a comparison result.

In step 406, the first one of the uplink grants is selected using the comparison result by the MAC layer. In the example in FIG. 6, in step 406, because the second PHY priority of the CG $CG_{402}(1)$ is higher than the first PHY priority of the DG $DG_{402}(0)$ in the comparison result, according to rule (1) of the PHY-based prioritization in step 302 described with reference to FIG. 3, the MAC layer MAC_{400} selects the CG $CG_{402}(1)$ (i.e., the CG $CG_{402}(1)$ is the first one of the uplink grants).

In step 408, only the first MAC PDU for the first one of the uplink grants is generated by the MAC layer. In the example in FIG. 6, in step 408, the MAC layer MAC_{400} performs a step S_{408} of encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH3 and an LCH LCH4 into a first MAC PDU for the CG $CG_{402}(1)$ and not encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH1 and an LCH LCH2 into a second MAC PDU for the DG $DG_{402}(0)$. Thus, even when the LCH-based prioritization is not configured in step 302, only the first MAC PDU for the CG $CG_{402}(1)$ is generated. Examples of the LCP mapping restrictions have been described with reference to step 108 in FIG. 1 and are omitted here.

In step 312, the first MAC PDU is delivered by the MAC layer to the PHY layer. In the example in FIG. 6, in step 312,

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the MAC layer MAC_{400} performs a step S_{312} of delivering the first MAC PDU for the CG $CG_{402}(1)$ to the PHY layer PHY_{400} . The CG $CG_{402}(1)$ of which the first MAC PDU is received by the PHY layer PHY_{400} is denoted as $CG_{312}(1)$. Because the second MAC PDU for the DG $DG_{402}(0)$ is not generated, the second MAC PDU for the DG $DG_{402}(0)$ is not delivered by the MAC layer MAC_{400} to the PHY layer PHY_{400} compared with the modified existing method 100 that delivers the second MAC PDU for the CG CG_{106} by the MAC layer MAC_{100} to the PHY layer PHY_{100} in step 110 which results in the second MAC PDU for the CG $CG_{110}(0)$ in the PHY layer PHY_{100} .

In step 314, the first MAC PDU is uplink transmitted by the PHY layer. In the example in FIG. 6, the PHY layer PHY_{400} performs a step S_{314} of uplink transmitting the first MAC PDU for the CG $CG_{312}(1)$. Because the second MAC PDU for the DG $DG_{402}(0)$ is not generated, the deficiency of the loss of the second MAC PDU, the inefficiency due to the need for retransmission, or the inefficiency due to the need for cancellation of the second MAC PDU is improved.

Referring to FIGS. 4 and 7, FIG. 4 is a flowchart illustrating a method 400 by a UE for PHY-based prioritization without MAC LCH-based prioritization, wherein PHY priority comparison for the PHY-based prioritization is performed by a MAC layer, in accordance with some embodiments of the present disclosure. FIG. 7 is a schematic diagram illustrating an example of applying the method 400 to an overlapping conflict in time domain between a DG with a high PHY priority and a CG with a high PHY priority. The method 400 includes the following steps.

In step 302, MAC LCH-based prioritization is not configured and PHY-based prioritization is configured by a UE. The step 302 has been described with reference to step 302 in FIG. 3 and is omitted here.

In step 304, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively, is received by a PHY layer of the UE. Illustratively, in the example in FIG. 7, the first uplink grant is a DG with a high PHY priority $DG_{304}(1)$ (i.e., with a priority index of 1 for the first PHY priority). The second uplink grant is a CG with a high PHY priority $CG_{304}(1)$ (i.e., with a priority index of 1 for the second PHY priority). The DG $DG_{304}(1)$ and the CG $CG_{304}(1)$ have an overlapping conflict in time domain, i.e., a PUSCH duration of the DG $DG_{304}(1)$ overlaps a PUSCH duration of the CG $CG_{304}(1)$. In step 304, a PHY layer PHY_{400} of the UE performs a step S_{304} of receiving the DG $DG_{304}(1)$ and the CG $CG_{304}(1)$.

In step 306, whether uplink transmission for the uplink grants is not on-going is determined by the PHY layer. The step 306 has been described with reference to step 306 in FIG. 3 and is omitted here.

In step 402, first information is delivered by the PHY layer to a MAC layer of the UE, wherein the first information includes the first uplink grant and the second uplink grant, and the first PHY priority and the second PHY priority. Step 402 is an embodiment of the step 308. In this embodiment, the first information explicitly indicates the first PHY priority and the second PHY priority. In the example in FIG. 7, in step 402, the PHY layer PHY_{400} performs a step S_{402} of delivering first information that includes the DG with the high PHY priority $DG_{304}(1)$ and the CG with the high PHY priority $CG_{304}(1)$ to a MAC layer MAC_{400} of the UE. The DG with the high PHY priority in the MAC layer MAC_{400} has the same content as the DG with the high PHY priority $DG_{304}(1)$ in the PHY layer PHY_{400} and is denoted as $DG_{402}(1)$. The CG with the high PHY

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priority in the MAC layer MAC_{400} has the same content as the CG with the high PHY priority $CG_{304}(1)$ is denoted as $CG_{402}(1)$.

The step 310 in FIG. 3 includes the steps 404, 406, and 408 in FIG. 4.

In step 404, the first PHY priority is compared with the second PHY priority to obtain a comparison result by the MAC layer. In the example in FIG. 7, in step 404, the MAC layer MAC_{400} compares the first PHY priority of the DG with the high PHY priority $DG_{402}(1)$ and the second PHY priority of the CG with the high PHY priority $CG_{402}(1)$ to obtain that the first PHY priority of the DG $DG_{402}(1)$ is the same as the second PHY priority of the CG $CG_{402}(1)$ as a comparison result.

In step 406, the first one of the uplink grants is selected using the comparison result by the MAC layer. In the example in FIG. 7, in step 406, because the first PHY priority of the DG $DG_{402}(1)$ is the same as the second PHY priority of the CG $CG_{402}(1)$ in the comparison result, according to rule (4) of the PHY-based prioritization in step 302 described with reference to FIG. 3, the MAC layer MAC_{400} selects a predetermined one of the uplink grants which is the DG $DG_{402}(1)$ (i.e., the DG $DG_{402}(1)$ is the first one of the uplink grants).

In step 408, only the first MAC PDU for the first one of the uplink grants is generated by the MAC layer. In the example in FIG. 7, in step 408, the MAC layer MAC_{400} performs a step S_{408} of encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH1 and an LCH LCH2 into a first MAC PDU for the DG $DG_{402}(1)$ and not encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH3 and an LCH LCH4 into a second MAC PDU for the CG $CG_{402}(1)$. Thus, even when the LCH-based prioritization is not configured in step 302, only the first MAC PDU for the DG $DG_{402}(1)$ is generated. Examples of the LCP mapping restrictions have been described with reference to step 108 in FIG. 1 and are omitted here.

In step 312, the first MAC PDU is delivered by the MAC layer to the PHY layer. In the example in FIG. 7, in step 312, the MAC layer MAC_{400} performs a step S_{312} of delivering the first MAC PDU for the DG $DG_{402}(1)$ to the PHY layer PHY_{400} . The DG $DG_{402}(1)$ of which the first MAC PDU is received by the PHY layer PHY_{400} is denoted as $DG_{312}(1)$. Because the second MAC PDU for the CG $CG_{402}(1)$ is not generated, the second MAC PDU for the CG $CG_{402}(1)$ is not delivered by the MAC layer MAC_{400} to the PHY layer PHY_{400} compared with the modified existing method 100 that delivers the second MAC PDU for the CG CG_{106} by the MAC layer MAC_{100} to the PHY layer PHY_{100} in step 110 which results in the second MAC PDU for the CG $CG_{110}(1)$ in the PHY layer PHY_{100} .

In step 314, the first MAC PDU is uplink transmitted by the PHY layer. In the example in FIG. 7, the PHY layer PHY_{400} performs a step S_{314} of uplink transmitting the first MAC PDU for the DG $DG_{402}(1)$. Because the second MAC PDU for the CG $DG_{402}(1)$ is not generated, the deficiency of the loss of the second MAC PDU or the inefficiency due to the need for cancellation of the second MAC PDU is improved.

Referring to FIGS. 4 and 8, FIG. 4 is a flowchart illustrating a method 400 by a UE for PHY-based prioritization without MAC LCH-based prioritization, wherein PHY priority comparison for the PHY-based prioritization is performed by a MAC layer, in accordance with some embodiments of the present disclosure. FIG. 8 is a schematic diagram illustrating an example of applying the method 400

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to an overlapping conflict in time domain between a DG with a low PHY priority and a CG with a low PHY priority. The method 400 includes the following steps.

In step 302, MAC LCH-based prioritization is not configured and PHY-based prioritization is configured by a UE. The step 302 has been described with reference to step 302 in FIG. 3 and is omitted here.

In step 304, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively, is received by a PHY layer of the UE. Illustratively, in the example in FIG. 8, the first uplink grant is a DG with a low PHY priority $DG_{304}(0)$ (i.e., with a priority index of 0 for the first PHY priority). The second uplink grant is a CG with a low PHY priority $CG_{304}(0)$ (i.e., with a priority index of 0 for the second PHY priority). The DG $DG_{304}(0)$ and the CG $CG_{304}(0)$ have an overlapping conflict in time domain, i.e., a PUSCH duration of the DG $DG_{304}(0)$ overlaps a PUSCH duration of the CG $CG_{304}(0)$. In step 304, a PHY layer PHY_{400} of the UE performs a step S_{304} of receiving the DG $DG_{304}(0)$ and the CG $CG_{304}(0)$.

In step 306, whether uplink transmission for the uplink grants is not on-going is determined by the PHY layer. The step 306 has been described with reference to step 306 in FIG. 3 and is omitted here.

In step 402, first information is delivered by the PHY layer to a MAC layer of the UE, wherein the first information includes the first uplink grant and the second uplink grant, and the first PHY priority and the second PHY priority. Step 402 is an embodiment of the step 308. In this embodiment, the first information explicitly indicates the first PHY priority and the second PHY priority. In the example in FIG. 8, in step 402, the PHY layer PHY_{400} performs a step S_{402} of delivering first information that includes the DG with the low PHY priority $DG_{304}(0)$ and the CG with the low PHY priority $CG_{304}(0)$ to a MAC layer MAC_{400} of the UE. The DG with the low PHY priority in the MAC layer MAC_{400} has the same content as the DG with the low PHY priority $DG_{304}(0)$ in the PHY layer PHY_{400} and is denoted as $DG_{402}(0)$. The CG with the low PHY priority in the MAC layer MAC_{400} has the same content as the CG with the low PHY priority $CG_{304}(0)$ is denoted as $CG_{402}(0)$.

The step 310 in FIG. 3 includes the steps 404, 406, and 408 in FIG. 4.

In step 404, the first PHY priority is compared with the second PHY priority to obtain a comparison result by the MAC layer. In the example in FIG. 8, in step 404, the MAC layer MAC_{400} compares the first PHY priority of the DG with the low PHY priority $DG_{402}(0)$ and the second PHY priority of the CG with the low PHY priority $CG_{402}(0)$ to obtain that the first PHY priority of the DG $DG_{402}(0)$ is the same as the second PHY priority of the CG $CG_{402}(0)$ as a comparison result.

In step 406, the first one of the uplink grants is selected using the comparison result by the MAC layer. In the example in FIG. 8, in step 406, because the first PHY priority of the DG $DG_{402}(0)$ is the same as the second PHY priority of the CG $CG_{402}(0)$ in the comparison result, according to rule (4) of the PHY-based prioritization in step 302 described with reference to FIG. 3, the MAC layer MAC_{400} selects a predetermined one of the uplink grants which is the DG $DG_{402}(0)$ (i.e., the DG $DG_{402}(0)$ is the first one of the uplink grants).

In step 408, only the first MAC PDU for the first one of the uplink grants is generated by the MAC layer. In the example in FIG. 8, in step 408, the MAC layer MAC_{400} performs a step S_{408} of encapsulating, on the basis of LCP

mapping restrictions, data from an LCH LCH1 and an LCH LCH2 into a first MAC PDU for the DG DG₄₀₂(0) and not encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH3 and an LCH LCH4 into a second MAC PDU for the CG CG₄₀₂(0). Thus, even when the LCH-based prioritization is not configured in step 302, only the first MAC PDU for the DG DG₄₀₂(0) is generated. Examples of the LCP mapping restrictions have been described with reference to step 108 in FIG. 1 and are omitted here.

In step 312, the first MAC PDU is delivered by the MAC layer to the PHY layer. In the example in FIG. 8, in step 312, the MAC layer MAC₄₀₀ performs a step S₃₁₂ of delivering the first MAC PDU for the DG DG₄₀₂(0) to the PHY layer PHY₄₀₀. The DG DG₄₀₂(0) of which the first MAC PDU is received by the PHY layer PHY₄₀₀ is denoted as DG₃₁₂(0). Because the second MAC PDU for the CG CG₄₀₂(0) is not generated, the second MAC PDU for the CG CG₄₀₂(0) is not delivered by the MAC layer MAC₄₀₀ to the PHY layer PHY₄₀₀ compared with the modified existing method 100 that delivers the second MAC PDU for the CG CG₁₀₆ by the MAC layer MAC₁₀₀ to the PHY layer PHY₁₀₀ in step 110 which results in the second MAC PDU for the CG CG₁₁₀(0) in the PHY layer PHY₁₀₀.

In step 314, the first MAC PDU is uplink transmitted by the PHY layer. In the example in FIG. 8, the PHY layer PHY₄₀₀ performs a step S₃₁₄ of uplink transmitting the first MAC PDU for the DG DG₄₀₂(0). Because the second MAC PDU for the CG CG₄₀₂(0) is not generated, the deficiency of the loss of the second MAC PDU or the inefficiency due to the need for cancellation of the second MAC PDU is improved.

Referring to FIGS. 9 and 10, FIG. 9 is a flowchart illustrating a method 900 by a UE for PHY-based prioritization without MAC LCH-based prioritization, wherein PHY priority comparison for the PHY-based prioritization is performed by a PHY layer, in accordance with some embodiments of the present disclosure. FIG. 10 is a schematic diagram illustrating an example of applying the method 900 to an overlapping conflict in time domain between a DG with a high PHY priority and a CG with a low PHY priority. The method 900 includes the following steps.

In step 302, MAC LCH-based prioritization is not configured and PHY-based prioritization is configured by a UE. The step 302 has been described with reference to step 302 in FIG. 3 and is omitted here.

In step 304, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively, is received by a PHY layer of the UE. Illustratively, in the example in FIG. 10, the first uplink grant is a DG with a high PHY priority DG₃₀₄(1) (i.e., with a priority index of 1 for the first PHY priority). The second uplink grant is a CG with a low PHY priority CG₃₀₄(0) (i.e., with a priority index of 0 for the second PHY priority). The DG DG₃₀₄(1) and the CG CG₃₀₄(0) have an overlapping conflict in time domain, i.e., a PUSCH duration of the DG DG₃₀₄(1) overlaps a PUSCH duration of the CG CG₃₀₄(0). In step 304, a PHY layer PHY₉₀₀ of the UE performs a step S₃₀₄ of receiving the DG DG₃₀₄(1) and the CG CG₃₀₄(0).

In step 306, whether uplink transmission for the uplink grants is not on-going is determined by the PHY layer. The step 306 has been described with reference to step 306 in FIG. 3 and is omitted here.

Steps 902 to 906 are further included by the method 300 in FIG. 3 between steps 306 and 308. The step 906 is an embodiment of the step 308 in FIG. 3.

In step 902, the first PHY priority is compared with the second PHY priority to obtain a comparison result by the PHY layer. In the example in FIG. 10, in step 902, the PHY layer PHY₉₀₀ compares the first PHY priority of the DG with the high PHY priority DG₃₀₄(1) and the second PHY priority of the CG with the low PHY priority CG₃₀₄(0) to obtain that the first PHY priority of the DG DG₃₀₄(1) is higher than the second PHY priority of the CG CG₃₀₄(0) as a comparison result.

In step 904, the first one of the uplink grants is selected using the comparison result by the PHY layer. In the example in FIG. 10, in step 904, because the first PHY priority of the DG DG₃₀₄(1) is higher than the second PHY priority of the CG CG₃₀₄(0) in the comparison result, according to rule (3) of the PHY-based prioritization in step 302 described with reference to FIG. 3, the PHY layer PHY₉₀₀ selects the DG DG₃₀₄(1) (i.e., the DG DG₃₀₄(1) is the first one of the uplink grants).

In step 906, first information is delivered by the PHY layer to a MAC layer of the UE, wherein the first information includes the first one of the uplink grants, but does not include the other one of the uplink grants. As mentioned above, step 906 is an embodiment of the step 308. In this embodiment, the first information implicitly indicates the first PHY priority and the second PHY priority. In the example in FIG. 10, in step 906, the PHY layer PHY₉₀₀ performs a step S₉₀₆ of delivering first information that includes the DG without the first PHY priority DG₉₀₆ but does not include the CG with the second PHY priority CG₃₀₄(0) or the CG without the second PHY priority CG₃₀₄ to a MAC layer MAC₉₀₀ of the UE. The DG DG₉₀₆ has the same content as the DG DG₃₀₄(1) except the DG DG₉₀₆ does not have information of the first PHY priority.

In step 310, only the first MAC PDU for the first one of the uplink grants is generated by the MAC layer. In the example in FIG. 10, in step 310, the MAC layer MAC₉₀₀ performs a step S₃₁₀ of encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH1 and an LCH LCH2 into a first MAC PDU for the DG DG₉₀₆. Because the first information does not include the CG CG₃₀₄(0) or CG₃₀₄, the step S₃₁₀ further includes not encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH3 and an LCH LCH4 into a second MAC PDU for the CG CG₃₀₄(0) or CG₃₀₄. Thus, even when the LCH-based prioritization is not configured in step 302, only the first MAC PDU for the DG DG₉₀₆ is generated. Examples of the LCP mapping restrictions have been described with reference to step 108 in FIG. 1 and are omitted here.

In step 312, the first MAC PDU is delivered by the MAC layer to the PHY layer. In the example in FIG. 10, in step 312, the MAC layer MAC₉₀₀ performs a step S₃₁₂ of delivering the first MAC PDU for the DG DG₉₀₆ to the PHY layer PHY₉₀₀. The DG DG₉₀₆ of which the first MAC PDU is received by the PHY layer PHY₉₀₀ is denoted as DG₃₁₂. Because the second MAC PDU for the CG CG₃₀₄(0) or CG₃₀₄ is not generated, the second MAC PDU for the CG CG₃₀₄(0) or CG₃₀₄ is not delivered by the MAC layer MAC₉₀₀ to the PHY layer PHY₉₀₀ compared with the modified existing method 100 that delivers the second MAC PDU for the CG CG₁₀₆ by the MAC layer MAC₁₀₀ to the PHY layer PHY₁₀₀ in step 110 which results in the second MAC PDU for the CG CG₁₁₀(0) in the PHY layer PHY₁₀₀.

In step 314, the first MAC PDU is uplink transmitted by the PHY layer. In the example in FIG. 10, the PHY layer PHY₉₀₀ performs a step S₃₁₄ of uplink transmitting the first MAC PDU for the DG DG₃₁₂. Because the second MAC PDU for the CG CG₃₀₄(0) or CG₃₀₄ is not generated, the

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deficiency of the loss of the second MAC PDU or the inefficiency due to the need for cancellation of the second MAC PDU is improved.

Referring to FIGS. 9 and 11, FIG. 9 is a flowchart illustrating a method 900 by a UE for PHY-based prioritization without MAC LCH-based prioritization, wherein PHY priority comparison for the PHY-based prioritization is performed by a PHY layer, in accordance with some embodiments of the present disclosure. FIG. 11 is a schematic diagram illustrating an example of applying the method 900 to an overlapping conflict in time domain between a DG with a low PHY priority and a CG with a high PHY priority. The method 900 includes the following steps.

In step 302, MAC LCH-based prioritization is not configured and PHY-based prioritization is configured by a UE. The step 302 has been described with reference to step 302 in FIG. 3 and is omitted here.

In step 304, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively, is received by a PHY layer of the UE. Illustratively, in the example in FIG. 11, the first uplink grant is a DG with a low PHY priority $DG_{304}(0)$ (i.e., with a priority index of 0 for the first PHY priority). The second uplink grant is a CG with a high PHY priority $CG_{304}(1)$ (i.e., with a priority index of 1 for the second PHY priority). The DG $DG_{304}(0)$ and the CG $CG_{304}(1)$ have an overlapping conflict in time domain, i.e., a PUSCH duration of the DG $DG_{304}(0)$ overlaps a PUSCH duration of the CG $CG_{304}(1)$. In step 304, a PHY layer PHY_{900} of the UE performs a step S_{304} of receiving the DG $DG_{304}(0)$ and the CG $CG_{304}(1)$.

In step 306, whether uplink transmission for the uplink grants is not on-going is determined by the PHY layer. The step 306 has been described with reference to step 306 in FIG. 3 and is omitted here.

Steps 902 to 906 are further included by the method 300 in FIG. 3 between steps 306 and 308. The step 906 is an embodiment of the step 308 in FIG. 3.

In step 902, the first PHY priority is compared with the second PHY priority to obtain a comparison result by the PHY layer. In the example in FIG. 11, in step 902, the PHY layer PHY_{900} compares the first PHY priority of the DG with the low PHY priority $DG_{304}(0)$ and the second PHY priority of the CG with the high PHY priority $CG_{304}(1)$ to obtain that the second PHY priority of the CG $CG_{304}(1)$ is higher than the first PHY priority of the DG $DG_{304}(0)$ as a comparison result.

In step 904, the first one of the uplink grants is selected using the comparison result by the PHY layer. In the example in FIG. 11, in step 904, because the second PHY priority of the CG $CG_{304}(1)$ is higher than the first PHY priority of the DG $DG_{304}(0)$ in the comparison result, according to rule (1) of the PHY-based prioritization in step 302 described with reference to FIG. 3, the PHY layer PHY_{900} selects the CG $CG_{304}(1)$ (i.e., the CG $CG_{304}(1)$ is the first one of the uplink grants).

In step 906, first information is delivered by the PHY layer to a MAC layer of the UE, wherein the first information includes the first one of the uplink grants, but does not include the other one of the uplink grants. As mentioned above, step 906 is an embodiment of the step 308. In this embodiment, the first information implicitly indicates the first PHY priority and the second PHY priority. In the example in FIG. 11, in step 906, the PHY layer PHY_{900} performs a step S_{906} of delivering first information that includes the CG without the second PHY priority CG_{906} but does not include the DG with the first PHY priority DG_{304}

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(0) or the DG without the first PHY priority DG_{304} to a MAC layer MAC_{900} of the UE. The CG CG_{906} has the same content as the CG $CG_{304}(1)$ except the CG CG_{906} does not have information of the second PHY priority.

In step 310, only the first MAC PDU for the first one of the uplink grants is generated by the MAC layer. In the example in FIG. 11, in step 310, the MAC layer MAC_{900} performs a step S_{310} of encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH1 and an LCH LCH2 into a first MAC PDU for the CG CG_{906} . Because the first information does not include the DG $DG_{304}(0)$ or DG_{304} , the step S_{310} further includes not encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH3 and an LCH LCH4 into a second MAC PDU for the DG $DG_{304}(0)$ or DG_{304} . Thus, even when the LCH-based prioritization is not configured in step 302, only the first MAC PDU for the CG CG_{906} is generated. Examples of the LCP mapping restrictions have been described with reference to step 108 in FIG. 1 and are omitted here.

In step 312, the first MAC PDU is delivered by the MAC layer to the PHY layer. In the example in FIG. 11, in step 312, the MAC layer MAC_{900} performs a step S_{312} of delivering the first MAC PDU for the CG CG_{906} to the PHY layer PHY_{900} . The CG CG_{906} of which the first MAC PDU is received by the PHY layer PHY_{900} is denoted as CG_{312} . Because the second MAC PDU for the DG $DG_{304}(0)$ or DG_{304} is not generated, the second MAC PDU for the DG $DG_{304}(0)$ or DG_{304} is not delivered by the MAC layer MAC_{900} to the PHY layer PHY_{900} compared with the modified existing method 100 that delivers the second MAC PDU for the CG CG_{106} by the MAC layer MAC_{100} to the PHY layer PHY_{100} in step 110 which results in the second MAC PDU for the CG $CG_{110}(0)$ in the PHY layer PHY_{100} .

In step 314, the first MAC PDU is uplink transmitted by the PHY layer. In the example in FIG. 11, the PHY layer PHY_{900} performs a step S_{314} of uplink transmitting the first MAC PDU for the CG CG_{312} . Because the second MAC PDU for the DG $DG_{304}(0)$ or DG_{304} is not generated, the deficiency of the loss of the second MAC PDU, the inefficiency due to the need for retransmission, or the inefficiency due to the need for cancellation of the second MAC PDU is improved.

Referring to FIGS. 9 and 12, FIG. 9 is a flowchart illustrating a method 900 by a UE for PHY-based prioritization without MAC LCH-based prioritization, wherein PHY priority comparison for the PHY-based prioritization is performed by a PHY layer, in accordance with some embodiments of the present disclosure. FIG. 12 is a schematic diagram illustrating an example of applying the method 900 to an overlapping conflict in time domain between a DG with a high PHY priority and a CG with a high PHY priority. The method 900 includes the following steps.

In step 302, MAC LCH-based prioritization is not configured and PHY-based prioritization is configured by a UE. The step 302 has been described with reference to step 302 in FIG. 3 and is omitted here.

In step 304, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively, is received by a PHY layer of the UE. Illustratively, in the example in FIG. 12, the first uplink grant is a DG with a high PHY priority $DG_{304}(1)$ (i.e., with a priority index of 1 for the first PHY priority). The second uplink grant is a CG with a high PHY priority $CG_{304}(1)$ (i.e., with a priority index of 1 for the second PHY priority). The DG $DG_{304}(1)$ and the CG $CG_{304}(1)$ have an overlapping conflict in time domain, i.e.,

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a PUSCH duration of the DG $DG_{304}(1)$ overlaps a PUSCH duration of the CG $CG_{304}(1)$. In step 304, a PHY layer PHY_{900} of the UE performs a step S_{304} of receiving the DG $DG_{304}(1)$ and the CG $CG_{304}(1)$.

In step 306, whether uplink transmission for the uplink grants is not on-going is determined by the PHY layer. The step 306 has been described with reference to step 306 in FIG. 3 and is omitted here.

Steps 902 to 906 are further included by the method 300 in FIG. 3 between steps 306 and 308. The step 906 is an embodiment of the step 308 in FIG. 3.

In step 902, the first PHY priority is compared with the second PHY priority to obtain a comparison result by the PHY layer. In the example in FIG. 12, in step 902, the PHY layer PHY_{900} compares the first PHY priority of the DG with the high PHY priority $DG_{304}(1)$ and the second PHY priority of the CG with the high PHY priority $CG_{304}(1)$ to obtain that the first PHY priority of the DG $DG_{304}(1)$ is the same as the second PHY priority of the CG $CG_{304}(1)$ as a comparison result.

In step 904, the first one of the uplink grants is selected using the comparison result by the PHY layer. In the example in FIG. 12, in step 904, because the first PHY priority of the DG $DG_{304}(1)$ is the same as the second PHY priority of the CG $CG_{304}(1)$ in the comparison result, according to rule (4) of the PHY-based prioritization in step 302 described with reference to FIG. 3, the PHY layer PHY_{900} selects a predetermined one of the uplink grants which is the DG $DG_{304}(1)$ (i.e., the DG $DG_{304}(1)$ is the first one of the uplink grants).

In step 906, first information is delivered by the PHY layer to a MAC layer of the UE, wherein the first information includes the first one of the uplink grants, but does not include the other one of the uplink grants. As mentioned above, step 906 is an embodiment of the step 308. In this embodiment, the first information implicitly indicates the first PHY priority and the second PHY priority. In the example in FIG. 12, in step 906, the PHY layer PHY_{900} performs a step S_{906} of delivering first information that includes the DG without the first PHY priority DG_{906} but does not include the CG with the second PHY priority $CG_{304}(1)$ or the CG without the second PHY priority CG_{304} to a MAC layer MAC_{900} of the UE. The DG DG_{906} has the same content as the DG $DG_{304}(1)$ except the DG DG_{906} does not have information of the first PHY priority.

In step 310, only the first MAC PDU for the first one of the uplink grants is generated by the MAC layer. In the example in FIG. 12, in step 310, the MAC layer MAC_{900} performs a step S_{310} of encapsulating, on the basis of LCP mapping restrictions, data from an LCH $LCH1$ and an LCH $LCH2$ into a first MAC PDU for the DG DG_{906} . Because the first information does not include the CG $CG_{304}(1)$ or CG_{304} , the step S_{310} further includes not encapsulating, on the basis of LCP mapping restrictions, data from an LCH $LCH3$ and an LCH $LCH4$ into a second MAC PDU for the CG $CG_{304}(1)$ or CG_{304} . Thus, even when the LCH-based prioritization is not configured in step 302, only the first MAC PDU for the DG DG_{906} is generated. Examples of the LCP mapping restrictions have been described with reference to step 108 in FIG. 1 and are omitted here.

In step 312, the first MAC PDU is delivered by the MAC layer to the PHY layer. In the example in FIG. 12, in step 312, the MAC layer MAC_{900} performs a step S_{312} of delivering the first MAC PDU for the DG DG_{906} to the PHY layer PHY_{900} . The DG DG_{906} of which the first MAC PDU is received by the PHY layer PHY_{900} is denoted as DG_{312} . Because the second MAC PDU for the CG $CG_{304}(1)$ or

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CG_{304} is not generated, the second MAC PDU for the CG $CG_{304}(1)$ or CG_{304} is not delivered by the MAC layer MAC_{900} to the PHY layer PHY_{900} compared with the modified existing method 100 that delivers the second MAC PDU for the CG CG_{106} by the MAC layer MAC_{100} to the PHY layer PHY_{100} in step 110 which results in the second MAC PDU for the CG $CG_{110}(0)$ in the PHY layer PHY_{100} .

In step 314, the first MAC PDU is uplink transmitted by the PHY layer. In the example in FIG. 12, the PHY layer PHY_{900} performs a step S_{314} of uplink transmitting the first MAC PDU for the DG DG_{312} . Because the second MAC PDU for the CG $CG_{304}(1)$ or CG_{304} is not generated, the deficiency of the loss of the second MAC PDU or the inefficiency due to the need for cancellation of the second MAC PDU is improved.

Referring to FIGS. 9 and 13, FIG. 9 is a flowchart illustrating a method 900 by a UE for PHY-based prioritization without MAC LCH-based prioritization, wherein PHY priority comparison for the PHY-based prioritization is performed by a PHY layer, in accordance with some embodiments of the present disclosure. FIG. 13 is a schematic diagram illustrating an example of applying the method 900 to an overlapping conflict in time domain between a DG with a low PHY priority and a CG with a low PHY priority. The method 900 includes the following steps.

In step 302, MAC LCH-based prioritization is not configured and PHY-based prioritization is configured by a UE. The step 302 has been described with reference to step 302 in FIG. 3 and is omitted here.

In step 304, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively, is received by a PHY layer of the UE. Illustratively, in the example in FIG. 13, the first uplink grant is a DG with a low PHY priority $DG_{304}(0)$ (i.e., with a priority index of 0 for the first PHY priority). The second uplink grant is a CG with a low PHY priority $CG_{304}(0)$ (i.e., with a priority index of 0 for the second PHY priority). The DG $DG_{304}(0)$ and the CG $CG_{304}(0)$ have an overlapping conflict in time domain, i.e., a PUSCH duration of the DG $DG_{304}(0)$ overlaps a PUSCH duration of the CG $CG_{304}(0)$. In step 304, a PHY layer PHY_{900} of the UE performs a step S_{304} of receiving the DG $DG_{304}(0)$ and the CG $CG_{304}(0)$.

In step 306, whether uplink transmission for the uplink grants is not on-going is determined by the PHY layer. The step 306 has been described with reference to step 306 in FIG. 3 and is omitted here.

Steps 902 to 906 are further included by the method 300 in FIG. 3 between steps 306 and 308. The step 906 is an embodiment of the step 308 in FIG. 3.

In step 902, the first PHY priority is compared with the second PHY priority to obtain a comparison result by the PHY layer. In the example in FIG. 13, in step 902, the PHY layer PHY_{900} compares the first PHY priority of the DG with the low PHY priority $DG_{304}(0)$ and the second PHY priority of the CG with the low PHY priority $CG_{304}(0)$ to obtain that the first PHY priority of the DG $DG_{304}(0)$ is the same as the second PHY priority of the CG $CG_{304}(0)$ as a comparison result.

In step 904, the first one of the uplink grants is selected using the comparison result by the PHY layer. In the example in FIG. 13, in step 904, because the first PHY priority of the DG $DG_{304}(0)$ is the same as the second PHY priority of the CG $CG_{304}(0)$ in the comparison result, according to rule (4) of the PHY-based prioritization in step 302 described with reference to FIG. 3, the PHY layer

PHY₉₀₀ selects a predetermined one of the uplink grants which is the DG DG₃₀₄(0) (i.e., the DG DG₃₀₄(0) is the first one of the uplink grants).

In step 906, first information is delivered by the PHY layer to a MAC layer of the UE, wherein the first information includes the first one of the uplink grants, but does not include the other one of the uplink grants. As mentioned above, step 906 is an embodiment of the step 308. In this embodiment, the first information implicitly indicates the first PHY priority and the second PHY priority. In the example in FIG. 13, in step 906, the PHY layer PHY₉₀₀ performs a step S₉₀₆ of delivering first information that includes the DG without the first PHY priority DG₉₀₆ but does not include the CG with the second PHY priority CG₃₀₄(0) or the CG without the second PHY priority CG₃₀₄ to a MAC layer MAC₉₀₀ of the UE. The DG DG₉₀₆ has the same content as the DG DG₃₀₄(0) except the DG DG₉₀₆ does not have information of the first PHY priority.

In step 310, only the first MAC PDU for the first one of the uplink grants is generated by the MAC layer. In the example in FIG. 13, in step 310, the MAC layer MAC₉₀₀ performs a step S₃₁₀ of encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH1 and an LCH LCH2 into a first MAC PDU for the DG DG₉₀₆. Because the first information does not include the CG CG₃₀₄(0) or CG₃₀₄, the step S₃₁₀ further includes not encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH3 and an LCH LCH4 into a second MAC PDU for the CG CG₃₀₄(0) or CG₃₀₄. Thus, even when the LCH-based prioritization is not configured in step 302, only the first MAC PDU for the DG DG₉₀₆ is generated. Examples of the LCP mapping restrictions have been described with reference to step 108 in FIG. 1 and are omitted here.

In step 312, the first MAC PDU is delivered by the MAC layer to the PHY layer. In the example in FIG. 13, in step 312, the MAC layer MAC₉₀₀ performs a step S₃₁₂ of delivering the first MAC PDU for the DG DG₉₀₆ to the PHY layer PHY₉₀₀. The DG DG₉₀₆ of which the first MAC PDU is received by the PHY layer PHY₉₀₀ is denoted as DG₃₁₂. Because the second MAC PDU for the CG CG₃₀₄(0) or CG₃₀₄ is not generated, the second MAC PDU for the CG CG₃₀₄(0) or CG₃₀₄ is not delivered by the MAC layer MAC₉₀₀ to the PHY layer PHY₉₀₀ compared with the modified existing method 100 that delivers the second MAC PDU for the CG CG₁₀₆ by the MAC layer MAC₁₀₀ to the PHY layer PHY₁₀₀ in step 110 which results in the second MAC PDU for the CG CG₁₁₀(0) in the PHY layer PHY₁₀₀.

In step 314, the first MAC PDU is uplink transmitted by the PHY layer.

In the example in FIG. 13, the PHY layer PHY₉₀₀ performs a step S₃₁₄ of uplink transmitting the first MAC PDU for the DG DG₃₁₂. Because the second MAC PDU for the CG CG₃₀₄(0) or CG₃₀₄ is not generated, the deficiency of the loss of the second MAC PDU or the inefficiency due to the need for cancellation of the second MAC PDU is improved.

A high PHY priority has a priority index of 1 and a low PHY priority has a priority index of 0 is illustrative. Alternatively, the high PHY priority has a priority index of 0 and a low PHY priority index of 1.

An uplink grant and a PHY priority being integrated together and called an uplink grant with a high or low PHY priority (for example, a DG with a high PHY priority DG(1)) is illustrative. Alternatively, the uplink grant and the PHY priority are separated.

In step 906 in the embodiments described with reference to FIG. 9, the first information including the first one of the

uplink grants without the PHY priority is illustrative. Alternatively, the first information further includes the PHY priority of the first one of the uplink grants (e.g., the first information includes the first uplink grants with the PHY priority).

In the above examples, it is assumed that at least one LCH mapped to each of the uplink grants has data and thus, the first information needs to be used for generating only the first MAC PDU for the first one of the uplink grants. In some embodiments, it is possible that every LCH mapped to one of the uplink grants has no data. In this case, on the basis of the method 400 described with reference to FIG. 4, both the uplink grants are delivered by the PHY layer to the MAC layer in the step 402, and thus the MAC layer can select the other one of the uplink grants to which at least one LCH is mapped has data. Depending on implementation, comparison of the PHY priorities in the first information may be performed before or after determination of whether every LCH mapped to one of the uplink grants has no data, and the earlier performed step may prevent execution of the later performed step.

Referring to FIG. 14, FIG. 14 is a schematic diagram illustrating an example of applying the method 400 that is modified so that comparison of PHY priorities in first information is performed before determination of whether every LCH mapped to one of the uplink grants has no data (hereafter referred to as a method 400'). In the example in FIG. 14, the method 400' is applied to an overlapping conflict in time domain between a DG with a low PHY priority and a CG with a high PHY priority, and every LCH mapped to the CG has no data. The method 400' includes the following steps.

In step 302, MAC LCH-based prioritization is not configured and PHY-based prioritization is configured by a UE. The step 302 has been described with reference to step 302 in FIG. 3 and is omitted here.

In step 304, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively, is received by a PHY layer of the UE. Illustratively, in the example in FIG. 14, the first uplink grant is a DG with a low PHY priority DG₃₀₄(0) (i.e., with a priority index of 0 for the first PHY priority). The second uplink grant is a CG with a high PHY priority CG₃₀₄(1) (i.e., with a priority index of 1 for the second PHY priority). The DG DG₃₀₄(0) and the CG CG₃₀₄(1) have an overlapping conflict in time domain, i.e., a PUSCH duration of the DG DG₃₀₄(0) overlaps a PUSCH duration of the CG CG₃₀₄(1). In step 304, a PHY layer PHY₁₄₀₀ of the UE performs a step S₃₀₄ of receiving the DG DG₃₀₄(0) and the CG CG₃₀₄(1).

In step 306, whether uplink transmission for the uplink grants is not on-going is determined by the PHY layer. The step 306 has been described with reference to step 306 in FIG. 3 and is omitted here.

In step 402, first information is delivered by the PHY layer to a MAC layer of the UE, wherein the first information includes the first uplink grant and the second uplink grant, and the first PHY priority and the second PHY priority. Step 402 is an embodiment of the step 308. In this embodiment, the first information explicitly indicates the first PHY priority and the second PHY priority. In the example in FIG. 14, in step 402, the PHY layer PHY₁₄₀₀ performs a step S₄₀₂ of delivering first information that includes the DG with the low PHY priority DG₃₀₄(0) and the CG with the high PHY priority CG₃₀₄(1) to a MAC layer MAC₁₄₀₀ of the UE. The DG with the low PHY priority in the MAC layer MAC₁₄₀₀ has the same content as the DG

with the low PHY priority $DG_{304}(0)$ in the PHY layer PHY_{1400} and is denoted as $DG_{402}(0)$. The CG with the high PHY priority in the MAC layer MAC_{1400} has the same content as the CG with the high PHY priority $CG_{304}(1)$ is denoted as $CG_{402}(1)$.

In step **404**, the first PHY priority is compared with the second PHY priority to obtain a comparison result by the MAC layer. In the example in FIG. 14, in step **404**, the MAC layer MAC_{1400} compares the first PHY priority of the DG with the low PHY priority $DG_{402}(0)$ and the second PHY priority of the CG with the high PHY priority $CG_{402}(1)$ to obtain that the second PHY priority of the CG $CG_{402}(1)$ is higher than the first PHY priority of the DG $DG_{402}(0)$ as a comparison result.

In step **1402**, whether every LCH mapped to one of the uplink grants has no data is determined by the MAC layer, wherein the one of the uplink grants is a CG with a higher PHY priority in the comparison result. In the example in FIG. 14, in step **1402**, because it is the CG $CG_{402}(1)$ that has the higher PHY priority, whether every LCH LCH3' or LCH4' mapped to the CG $CG_{402}(1)$ has no data needs to be determined. In the example in FIG. 14, a result of this determination is true. Thus, step **1404** is performed.

In step **1404**, the other one of the uplink grants is selected. In the example in FIG. 14, in step **1404**, because both of the grants (the $DG_{304}(0)$ and the $CG_{304}(1)$) are delivered to the MAC layer MAC_{1400} in step **402**, the other one of the uplink grants (the $DG_{304}(0)$) can be selected. Further, because the other one of the uplink grants is the DG $DG_{402}(0)$ that is requested because of data awaiting transmission, the MAC layer MAC_{1400} selects the DG $DG_{402}(0)$.

In step **1406**, only a third MAC PDU is generated for the other one of the uplink grants by the MAC layer. In the example in FIG. 14, in step **1406**, the MAC layer MAC_{1400} performs a step S_{1406} of encapsulating, on the basis of LCP mapping restrictions, data from an LCH LCH1 and an LCH LCH2 into only a third MAC PDU for the DG $DG_{402}(0)$.

In step **1408**, the third MAC PDU is delivered by the MAC layer to the PHY layer. In the example in FIG. 14, in step **1408**, the MAC layer MAC_{1400} performs a step S_{1408} of delivering the third MAC PDU for the DG $DG_{1408}(0)$ to the PHY layer PHY_{1400} . The DG $DG_{1408}(0)$ of which the third MAC PDU is received by the PHY layer PHY_{1400} is denoted as $DG_{1408}(0)$.

In step **1410**, the third MAC PDU is uplink transmitted by the PHY layer. In the example in FIG. 14, the PHY layer PHY_{1400} performs a step S_{1410} of uplink transmitting the third MAC PDU for the DG $DG_{1408}(0)$.

In the above example, the CG has the high PHY priority and the DG has the low PHY priority. For other situations such as a DG having a high PHY priority and a CG having a low PHY priority, or PHY priorities of a DG and a CG being the same, because the comparison of the PHY priorities itself is enough to cause the DG to be selected, determination of whether every LCH mapped to one of the uplink grants has no data need not to be performed.

In the above embodiment, comparison of the PHY priorities in the first information is performed before determination of whether every LCH mapped to the one of the uplink grants has no data. Alternatively, comparison of the PHY priorities in the first information is performed after determination of whether every LCH mapped to one of the uplink grants has no data. In this case, in step **1402'**, whether every LCH mapped to one of the uplink grants has no data is determined by the MAC layer, wherein the one of the uplink grants is a CG. If so, step **404** of comparing the first PHY priority with the second PHY priority to obtain a

comparison result is not performed. The step **1404** is directly performed after the step **1402'**. If not, step **404** and subsequent steps in the method **400** described with reference to FIG. 4 are performed.

Because the RAN node does not know whether there is data for at least one LCH mapped to the $CG_{304}(1)$, the RAN node may have no idea the third MAC PDU corresponds which of the uplink grants (the $DG_{304}(0)$ and the $CG_{304}(1)$), and decodes the third MAC PDU in a trial-and-error manner. To resolve this inefficiency, in some embodiments, an uplink indication channel that indicates a type of the uplink grant corresponding to the third MAC PDU is further uplink transmitted by the PHY layer along with the third MAC PDU.

Referring to FIG. 15, FIG. 15 is a schematic timeline diagram illustrating the latest time for performing a step of comparing the first PHY priority (e.g., a high PHY priority of a DG) with the second PHY priority (e.g., a low PHY priority of a CG) by the MAC layer in the method **400** in FIG. 4 in accordance with some embodiments of the present disclosure.

As illustrated in FIG. 15, at time T_{304} , step **304** is performed. In the example in FIG. 15, the first uplink grant is a DG with a high PHY priority $DG_{1500}(1)$ (i.e., with a priority index of 1 for the first PHY priority) and the second uplink grant is a CG with a low PHY priority $CG_{1500}(0)$ (i.e., with a priority index of 0 for the second PHY priority). When the DG $DG_{1500}(1)$ and the CG $CG_{1500}(0)$ is received at time T_{304} , it is also known that the DG $DG_{1500}(1)$ has an earlier one of scheduled transmission time T_{EST} of the uplink grants (the DG $DG_{1500}(1)$ and the CG $CG_{1500}(0)$).

Then, at time T_{402} , step **402** is performed. Then, at time T_{404} , step **404** is performed. The latest time for performing the step **404** is a first period of time T_{proc1} before the earlier one of scheduled transmission time T_{EST} of the uplink grants (the DG $DG_{1500}(1)$ and the CG $CG_{1500}(0)$). Because step **404** is performed by the MAC layer and relies on the first information delivered by the PHY layer to the MAC layer, the first period of time T_{proc1} does not include time for performing step **402**. The first period of time T_{proc1} includes time for performing steps **404**, **406**, **408**, and **312**. The first period of time T_{proc1} further includes time for preparing to perform the step **314** in the PHY layer.

Then, at time T_{314} , step **314** of uplink transmitting the first MAC PDU for the DG $DG_{1500}(1)$ is performed. Time T_{314} is the same as the earlier one of scheduled transmission time T_{EST} .

Referring to FIG. 16, FIG. 16 is a schematic timeline diagram illustrating the latest time for performing a step of comparing the first PHY priority (e.g., a low PHY priority of a DG) with the second PHY priority (e.g., a high PHY priority of a CG) by the MAC layer in the method **400** in FIG. 4 in accordance with some embodiments of the present disclosure.

The example in FIG. 16 illustrates that for the first uplink grant being a DG with a low PHY priority $DG_{1600}(0)$ and the second uplink grant being a CG with a high PHY priority $CG_{1600}(1)$, even though in step **314**, it is the CG $CG_{1600}(1)$ that is transmitted, the latest time for performing the step **404** is still a first period of time T_{proc1} before an earlier one of scheduled transmission time T_{EST} of the uplink grants (the DG $DG_{1600}(0)$ and the CG $CG_{1600}(1)$). The earlier one of scheduled transmission time T_{EST} is scheduled transmission time for the DG $DG_{1600}(0)$. Time T_{314} at which step **314** is performed is later than the earlier one of scheduled transmission time T_{EST} .

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Referring to FIG. 17, FIG. 17 is a schematic timeline diagram illustrating the latest time for performing a step of comparing the first PHY priority (e.g., a high PHY priority of a DG) with the second PHY priority (e.g., a low PHY priority of a CG) by the PHY layer in the method 900 in FIG. 9 in accordance with some embodiments of the present disclosure.

As illustrated in FIG. 17, at time T_{304} , step 304 is performed. In the example in FIG. 17, the first uplink grant is a DG with a high PHY priority $DG_{1500}(1)$ (i.e., with a priority index of 1 for the first PHY priority) and the second uplink grant is a CG with a low PHY priority $CG_{1500}(0)$ (i.e., with a priority index of 0 for the second PHY priority). When the DG $DG_{1500}(1)$ and the CG $CG_{1500}(0)$ is received at time T_{304} , it is also known that the DG $DG_{1500}(1)$ has an earlier one of scheduled transmission time T_{EST} of the uplink grants (the DG $DG_{1500}(1)$ and the CG $CG_{1500}(0)$).

Then, at time T_{902} , step 902 is performed. Then, at time T_{906} , step 906 is performed. The latest time for performing the step 902 is a second period of time T_{proc2} before the earlier one of scheduled transmission time T_{EST} of the uplink grants (the DG $DG_{1500}(1)$ and the CG $CG_{1500}(0)$). Because step 902 is performed by the PHY layer and results in the first information delivered by the PHY layer to the MAC layer, the second period of time T_{proc2} includes time for performing step 906. The second period of time T_{proc2} includes time for performing steps 902, 904, 906, 310, and 312. The second period of time T_{proc2} further includes time for preparing to perform the step 314 in the PHY layer.

Then, at time T_{314} , step 314 of uplink transmitting the first MAC PDU for the DG $DG_{1500}(1)$ is performed. Time T_{314} is the same as the earlier one of scheduled transmission time T_{EST} .

Referring to FIG. 18, FIG. 18 is a schematic timeline diagram illustrating the latest time for performing a step of comparing the first PHY priority (e.g., a low PHY priority of a DG) with the second PHY priority (e.g., a high PHY priority of a CG) by the PHY layer in the method 900 in FIG. 9 in accordance with some embodiments of the present disclosure.

The example in FIG. 18 illustrates that for the first uplink grant being a DG with a low PHY priority $DG_{1600}(0)$ and the second uplink grant being a CG with a high PHY priority $CG_{1600}(1)$, even though in step 314, it is the CG $CG_{1600}(1)$ that is transmitted, the latest time for performing the step 902 is still a second period of time T_{proc2} before an earlier one of scheduled transmission time T_{EST} of the uplink grants (the DG $DG_{1600}(0)$ and the CG $CG_{1600}(1)$). The earlier one of scheduled transmission time T_{EST} is scheduled transmission time for the DG $DG_{1600}(0)$. Time T_{314} at which step 314 is performed is later than the earlier one of scheduled transmission time T_{EST} .

In the above examples, the number of bits of each of the PHY priorities is one bit (i.e., the priority index of each of the PHY priorities can be 0 or 1). Because there are, for example, sixteen possible levels for any LCH priority used by the LCH-based prioritization, and there are two possible levels for any PHY priority used by the PHY-based prioritization, the number of levels of any PHY priority is not sufficient for some priority differences between the LCHs to be differentiated. In some embodiments, the number of bits of each of the PHY priorities is increased to two to the number of bits of any LCH priority used by the LCH-based prioritization. Thus, for both the method 400 in FIG. 4 and the method 900 in FIG. 9, the comparison result (in step 404 or 902) has higher precision.

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Referring to FIG. 19, FIG. 19 is a schematic table diagram illustrating a PHY priority-to-LCH priority mapping 1900, wherein the mapping 1900 is one-to-one, in accordance with some embodiments of the present disclosure. As illustrated in FIG. 19, the number of bits of each of the PHY priorities is the same as the number of bits of any LCH priority used by the LCH-based prioritization, and thus the PHY priority-to-LCH priority mapping 1900 is one-to-one. The number of bits of each of the PHY priorities is exemplarily four and thus there are sixteen possible levels for each of the PHY priorities which correspond to sixteen possible levels for any LCH priority used by the LCH-based prioritization, respectively. Thus, for both the method 400 in FIG. 4 and the method 900 in FIG. 9, the comparison result (in step 404 or 902) has the highest precision.

Even when the number of bits of each of the PHY priorities is increased to two, the increased number of possible levels of each of the PHY priorities are sufficient for an LCH priority (e.g., the highest LCH priority) for control information of a first service, an LCH priority (e.g., the highest LCH priority) for data of the first service, an LCH priority (e.g., the highest LCH priority) for control information of a second service, and an LCH priority (e.g., the highest LCH priority) for data of the second service to be differentiable. The first service and the second service are simultaneously provided by the UE and would have a high PHY priority and a low PHY priority, respectively, if the number of bits of each PHY priorities had been one bit.

For example, the first service is an ultra-reliable low latency communications (URLLC) service. The second service is a web-browsing service. Assuming that control information of the URLLC service is served by a first DG, data of the URLLC service is served by a CG, control information of the web-browsing service is served by a third DG, data of the web-browsing service is served by a fourth DG, because the number of bits of each of the PHY priorities is two, PHY priorities of the first DG, the CG, the third DG, and the fourth DG, respectively, are in a descending order. Thus, the PHY priorities of the first DG, the CG, the third DG, and the fourth DG, respectively, one-to-one correspond to LCH priorities of the control information of the URLLC service, the data of the URLLC service, the control information of the web-browsing service, and the data of the web-browsing service, respectively.

In the embodiments above, the PHY-based prioritization is configured and the LCH-based prioritization is not configured. In order to further enhance UE capability versatility, another case that PHY-based prioritization is not configured and LCH-based prioritization is configured is also proposed to be allowed. Such configuration is also agreed upon in the RAN2#109bis-e meeting.

When the existing method is modified in such a way that PHY-based prioritization is not configured and LCH-based prioritization is configured in situations that PHY-based prioritization is originally used to resolve the overlapping conflict, a RAN node may have no idea a MAC PDU uplink transmitted to the RAN node corresponds to which of the two uplink grants (a DG and a CG), as illustrated by the following steps.

In a first step, LCH-based prioritization is configured and the PHY-based prioritization is not configured by a UE.

In a second step, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain is received by a PHY layer of the UE. The first uplink grant is a DG. The second uplink grant is a CG.

In a third step, the uplink grants are delivered by the PHY layer to a MAC layer of the UE.

In a fourth step, only a fourth MAC PDU is generated for one of the uplink grants on the basis of LCP mapping restrictions and LCH-based prioritization. The MAC layer may encapsulate, on the basis of the LCP mapping restrictions, data from corresponding LCHs into two virtual MAC PDUs for the two uplink grants, respectively, assuming that LCH(s) corresponding to the CG has data available. The MAC layer selects one of the two virtual MAC PDUs on the basis of LCH-based prioritization. Then, the fourth MAC PDU which is real is generated for the selected one of the two virtual MAC PDUs. For the other one of the two virtual MAC PDUs, data is not taken out from the corresponding LCH(s) to form a real MAC PDU.

In a fifth step, the fourth MAC PDU is delivered by the MAC layer to the PHY layer.

In a sixth step, the fourth MAC PDU is uplink transmitted by the PHY layer.

When the RAN node receives the fourth MAC PDU from the UE, because the RAN node may have no idea the uplink grant corresponding to the fourth MAC PDU is not selected on the basis of the PHY priorities of the uplink grants, the RAN node may estimate a priority of each of the uplink grants for the fourth MAC PDU in a trial and error manner. On the basis of the priority of each of the uplink grants, the uplink grant corresponding to the fourth MAC PDU is determined.

To resolve inefficiency of trial and error, in some embodiments, an uplink indication channel that indicates a type of the uplink grant corresponding to the fourth MAC PDU is further uplink transmitted by the PHY layer along with the fourth MAC PDU.

Referring to FIG. 20, FIG. 20 is a schematic diagram of a system 2000 including a UE 2002 that is capable of PHY-based prioritization without LCH-based prioritization and/or LCH-based prioritization without PHY-based prioritization in accordance with some embodiments of the present disclosure. The system 2000 includes a UE 2002 and a RAN node 2008. In some embodiments, the UE 2002 is a higher-tier device that targets increased throughput, increased processing capability and high-power computation. The higher-tier device is, for example, a mobile phone, a tablet, or a subscriber station for factory applications. In other embodiments, the UE 2002 is a lower-tier device or a mid-tier device that is suitable for applications that may not require the increased throughput, the increased processing capability, and the high-power computation for the higher-tier device. The lower-tier device or the mid-tier device is a wearable device, or a sensor. The RAN node 2008 is, for example, a next generation NodeB (gNB), a next generation evolved NodeB (ng-eNB), or a gateway station for factory applications.

In some embodiments, the UE 2002 includes at least one processor 2004 and a communication interface 2006, wherein the at least one processor 2004 are connected to the communication interface 2006.

Embodiments described herein may be implemented into the UE 2002 in the system 2000 using any suitably configured hardware and/or software. The at least one processor 2004 is configured to execute instructions to implement any of the methods in accordance with the embodiments described above to be performed.

The at least one processor 2004 may be implemented as a "processing system." Examples of processors include microprocessors, microcontrollers, graphics processing units (GPUs), central processing units (CPUs), application processors, digital signal processors (DSPs), reduced instruction set computing (RISC) processors, systems on a

chip (SoC), baseband processors, field programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. At least one processor in the processing system may execute software.

The functions implemented in software may be stored on or encoded as one or more instructions or code on a computer-readable medium. Computer-readable media includes computer storage media, which may also be referred to as a non-transitory computer-readable medium. The term non-transitory computer-readable medium excludes transitory signals. Storage media may be any available media that can be accessed by a computer. By way of example, and not limitation, such computer-readable media can comprise a random-access memory (RAM), a read-only memory (ROM), an electrically erasable programmable ROM (EEPROM), optical disk storage, magnetic disk storage, other magnetic storage devices, combinations of the aforementioned types of computer-readable media, or any other medium that can be used to store computer executable code in the form of instructions or data structures that can be accessed by a computer.

The communication interface 2006 is configured to engage in communication with a communication interface of the RAN node 2008 via a communication path that includes at least one wireless link 2010.

The RAN node 2008 is configured to configure the PHY priorities, schedule the uplink grants, and receive and decode the uplink MAC PDU for any of the methods in accordance with the embodiments described above to be performed by the UE 2002.

What is claimed is:

1. A method performed by a user equipment (UE), comprising:
 - when media access control (MAC) logical channel-based prioritization (LCH-based prioritization) is not configured, and physical layer-based prioritization (PHY-based prioritization) is configured in the UE, receiving, by a PHY layer of the UE, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively; and
 - when uplink transmission for the uplink grants is not on-going,
 - delivering, by the PHY layer, first information to a MAC layer of the UE, wherein the first information is for resolving the overlapping conflict by comparing the first PHY priority with the second PHY priority; and
 - generating, by the MAC layer, only a first MAC protocol data unit (PDU) for a first one of the uplink grants using the first information;
 - wherein the first information comprises the first uplink grant and the second uplink grant, and the first PHY priority and the second PHY priority; and
 - wherein the step of generating, by the MAC layer of the UE, only the first MAC PDU for the first one of uplink grants using the first information comprises:
 - comparing, by the MAC layer, the first PHY priority with the second PHY priority to obtain a comparison result;
 - selecting, by the MAC layer, the first one of the uplink grants using the comparison result; and

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generating, by the MAC layer, only the first MAC PDU for the first one of the uplink grants.

2. The method of Claim 1, wherein the step of generating, by the MAC layer of the UE, only the first MAC PDU for the first one of the uplink grants using the first information is performed when at least one corresponding logical channel mapped to each of the uplink grants has data.

3. The method of claim 2, wherein the method further comprises:

when every logical channel mapped to one of the uplink grants has no data, selecting, by the MAC layer, the other one of the uplink grants; and

generating, by the MAC layer, only a second MAC PDU for the other one of the uplink grants.

4. The method of claim 1,

wherein the first information comprises the first one of the uplink grants but does not comprise the other one of the uplink grants;

wherein when uplink transmission for the uplink grants is not on-going, before the step of delivering, the method further comprises:

comparing, by the PHY layer, the first PHY priority with the second PHY priority to obtain a comparison result; and

selecting, by the PHY layer, the first one of the uplink grants using the comparison result.

5. The method of claim 1,

wherein the latest time for performing the step of comparing the first PHY priority with the second priority by the MAC layer is a first period of time before an earlier one of scheduled transmission time of the uplink grants;

wherein the first period of time does not include time for performing the step of delivering and includes time for performing the step of generating; or

wherein the latest time for performing the step of comparing the first PHY priority with the second priority by the PHY layer is a second period of time before an earlier one of scheduled transmission time of the uplink grants;

wherein the second period of time comprises time for performing the step of delivering and time for performing the step of generating.

6. The method of claim 1, wherein the number of bits of each of the PHY priorities is more than one bit and fewer than or the same as the number of bits of a logical channel priority used by the MAC LCH-based prioritization.

7. The method of claim 1,

wherein when one of the PHY priorities is higher than the other of the PHY priorities, the first one of the uplink grants is one of the uplink grants having the one of the PHY priorities; and/or

wherein when the PHY priorities are same, the first one of the uplink grants is a predetermined one of the uplink grants.

8. A user equipment (UE), comprising:

at least one processor and a communication interface, wherein the at least one processor is connected to the communication interface; and

wherein the at least one processor is configured to execute instructions to implement steps comprising:

when media access control (MAC) logical channel-based prioritization (LCH-based prioritization) is not configured, and physical layer-based prioritization (PHY-based prioritization) is configured in the UE, receiving, by a PHY layer of the UE, a first uplink grant and a second uplink grant that have an

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overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively; and

when uplink transmission for the uplink grants is not on-going,

delivering, by the PHY layer, first information to a MAC layer of the UE, wherein the first information is for resolving the overlapping conflict by comparing the first PHY priority with the second PHY priority; and

generating, by the MAC layer, only a first MAC protocol data unit (PDU) for a first one of the uplink grants using the first information;

wherein the first information comprises the first uplink grant and the second uplink grant, and the first PHY priority and the second PHY priority; and

wherein the step of generating, by the MAC layer of the UE, only the first MAC PDU for the first one of uplink grants using the first information comprises:

comparing, by the MAC layer, the first PHY priority with the second PHY priority to obtain a comparison result;

selecting, by the MAC layer, the first one of the uplink grants using the comparison result; and

generating, by the MAC layer, only the first MAC PDU for the first one of the uplink grants.

9. The UE of Claim 8, wherein the step of generating, by the MAC layer of the UE, only the first MAC PDU for the first one of the uplink grants using the first information is performed when at least one corresponding logical channel mapped to each of the uplink grants has data.

10. The UE of claim 9,

wherein the method further comprises:

when every logical channel mapped to one of the uplink grants has no data, selecting, by the MAC layer, the other one of the uplink grants; and

generating, by the MAC layer, only a second MAC PDU for the other one of the uplink grants.

11. The UE of claim 8,

wherein the first information comprises the first one of the uplink grants but does not comprise the other one of the uplink grants;

wherein when uplink transmission for the uplink grants is not on-going, before the step of delivering, the method further comprises:

comparing, by the PHY layer, the first PHY priority with the second PHY priority to obtain a comparison result; and

selecting, by the PHY layer, the first one of the uplink grants using the comparison result.

12. The UE of claim 8,

wherein the latest time for performing the step of comparing the first PHY priority with the second priority by the MAC layer is a first period of time before an earlier one of scheduled transmission time of the uplink grants;

wherein the first period of time does not include time for performing the step of delivering and includes time for performing the step of generating; or

wherein the latest time for performing the step of comparing the first PHY priority with the second priority by the PHY layer is a second period of time before an earlier one of scheduled transmission time of the uplink grants;

wherein the second period of time comprises time for performing the step of delivering and time for performing the step of generating.

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13. The UE of claim 8, wherein the number of bits of each of the PHY priorities is more than one bit and fewer than or the same as the number of bits of a logical channel priority used by the MAC LCH-based prioritization.

14. The UE of claim 8,

wherein when one of the PHY priorities is higher than the other of the PHY priorities, the first one of the uplink grants is one of the uplink grants having the one of the PHY priorities; and/or

wherein when the PHY priorities are same, the first one of the uplink grants is a predetermined one of the uplink grants.

15. A non-transitory computer readable medium, having instructions stored therein, wherein the instructions are executable by a processor to perform steps comprising:

when media access control (MAC) logical channel-based prioritization (LCH-based prioritization) is not configured, and physical layer-based prioritization (PHY-based prioritization) is configured in the UE,

receiving, by a PHY layer of the UE, a first uplink grant and a second uplink grant that have an overlapping conflict in time domain, and have a first PHY priority and a second PHY priority, respectively; and

when uplink transmission for the uplink grants is not on-going,

delivering, by the PHY layer, first information to a MAC layer of the UE, wherein the first information is for resolving the overlapping conflict by comparing the first PHY priority with the second PHY priority; and

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generating, by the MAC layer, only a first MAC protocol data unit (PDU) for a first one of the uplink grants using the first information;

wherein the first information comprises the first uplink grant and the second uplink grant, and the first PHY priority and the second PHY priority; and

wherein the step of generating, by the MAC layer of the UE, only the first MAC PDU for the first one of uplink grants using the first information comprises:

comparing, by the MAC layer, the first PHY priority with the second PHY priority to obtain a comparison result;

selecting, by the MAC layer, the first one of the uplink grants using the comparison result; and

generating, by the MAC layer, only the first MAC PDU for the first one of the uplink grants.

16. The non-transitory computer readable medium of Claim 15, wherein the step of generating, by the MAC layer of the UE, only the first MAC PDU for the first one of the uplink grants using the first information is performed when at least one corresponding logical channel mapped to each of the uplink grants has data.

17. The non-transitory computer readable medium of claim 16,

wherein the method further comprises:

when every logical channel mapped to one of the uplink grants has no data, selecting, by the MAC layer, the other one of the uplink grants; and

generating, by the MAC layer, only a second MAC PDU for the other one of the uplink grants.

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